ONLINE SUPPLEMENTARY MATERIAL

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Supplementary Table 1 Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) checklist[1]

Section/topic	#	Checklist item	Reported in main text
		Title	
Title	1	Identify the report as a systematic review, meta-analysis, or both.	p. 1
		Abstract	
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	p. 2-3
		Introduction	
Rationale	3	Describe the rationale for the review in the context of what is already known.	p. 4-6
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	p. 5
		Methods	
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	p. 3, 6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	p. 5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	p. 6-7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	online supp. box 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	p. 6-7
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	p. 7-8 and online supp. box 2 and online supp. table 2
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	online supp. box 2 and online supp. table 2
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	p. 8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	p. 8
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	p. 8
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	p. 8
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	p. 8-9
		Results	
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	p. 9-10 and figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	p. 10 and online supp. tables 3-8
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome-level assessment (see Item 12).	p. 10 -11 and online supp. tables 9-10
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group and (b) effect estimates and confidence intervals, ideally with a forest plot.	p. 11-12, table 1, figure 2 and online supp. figures 1-14
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	p. 11-12, table 1, figure 2 and online supp. figures 1-14
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	p. 10-11 and online supp. tables 9-10

Section/topic	#	Checklist item	Reported in main text
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	p. 12-13, table 2, and online supp. tables 11-12
Discussion			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., health care providers, users, and policy makers).	p. 13-16
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review level (e.g., incomplete retrieval of identified research, reporting bias).	p. 16-17
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	p. 17
Funding			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	p. 18-19

Fig: figure; NA: not applicable; p: page(s); Supp: supplementary.

Supplementary Box 1 Search strategy used to identify studies describing gonorrhea epidemiology in infertile populations

PubMed (April 29, 2019)

("Neisseria gonorrhoeae"[Mesh] OR "Gonorrhea"[Mesh] OR "Pelvic Inflammatory Disease"[Mesh] OR "Epididymitis"[Mesh] OR "Orchitis"[Mesh] OR "Seminal vesicle disease"[Mesh] OR "Neisseria gonorrhoeae"[Text] OR "Gonorrhoeae"[Text] OR "Gonorrhea"[Text] OR "Gonococcus"[Text] OR "Gonococci"[Text] OR "Gonococcal"[Text] OR "Gonococcal infection"[Text] OR "Pelvic inflammatory disease"[Text] OR "Gonococcal epididymitis"[Text] OR "Orchi-epididymitis"[Text] OR "Orchiepididymitis"[Text] OR "seminal disease"[Text] OR "seminal vesicle disease"[Text] OR "Seminal vasculitis"[Text]) AND ("Infertility"[Mesh] OR "Fertility"[Mesh] OR "Reproductive Techniques, Assisted"[Mesh] OR "Infertility"[Text] OR "Infertile"[Text] OR "Fertility"[Text] OR "Reproductive"[Text] OR "Subfertility"[Text] OR "Subfertile"[Text] OR "Sub-fertility"[Text] OR "Sub-fertile"[Text])

Embase (April 29, 2019)

(exp gonorrhea / or exp neisseria gonorrhoeae / or exp epididymitis / or exp orchitis / or exp pelvic inflammatory disease/ or gonorrhea.mp. or neisseria gonorrhoeae.mp. or gonorrhoeae.mp. or gonococcus.mp. or gonococci.mp. or gonococcal.mp. or gonococcal infection.mp. or pelvic inflammatory disease.mp. or gonococcal epididymitis.mp. or orchi-epididymitis.mp. or orchiepididymitis.mp. or seminal vesicle disease.mp. or seminal disease.mp. or seminal vasculitis.mp.) AND (exp infertility/ or exp fertility/ or exp infertility therapy/ or exp reproductive procedure/ or reproductive.mp. or infertility.mp. or infertile.mp. or fertility.mp. or subfertility.mp. or subfertile.mp. or sub-fertility.mp. or sub-fertile.mp.)

World Health Organization African Index Medicus (February 5, 2019)

Gonorrhea, gonorrhoeae, gonococcus, gonococci, gonococcal

World Health Organization Index Medicus for the Eastern Mediterranean (February 2, 2019)

Gonorrhea, gonorrhoeae, gonococcus, gonococci, gonococcal

World Health Organization Index Medicus for the South-East Asian (February 2, 2019)

Gonorrhea, gonorrhoeae, gonococcus, gonococci, gonococcal

World Health Organization Index Medicus for the Western Pacific (February 5, 2019)

Gonorrhea, gonorrhoeae, gonococcus, gonococci, gonococcal

Supplementary Box 2 Study extraction list

Publication-related variables

Author(s)

Publication year

Full citation

Study-related variables

Country

World Health organization (WHO) region (African Region, Region of the Americas, Eastern Mediterranean Region, European Region, South-East Asia Region, Western Pacific Region)

Year of data collection start and end

Study site

Study design

Sampling methodology (probability-based, non-probability-based)

Study population and its characteristics

- Sex
- Age
- Infertility type (primary, secondary, unclear)
- Infertility diagnosis (general or unclear, tubal factor, male factor, ovarian and non-tubal, mixed and unexplained)
- Presence of urogenital signs and symptoms (asymptomatic, symptomatic, unclear)

Gonorrhea infection

- Biological specimen type (urethral, vaginal, endocervical, serum, urine, semen)
- Sample size of tested population
- Number of participants positive for gonorrhea infection
- Reported gonorrhea prevalence
- Type of assay used for gonorrhea infection ascertainment (nucleic acid amplification test, culture, gram stain, direct fluorescent antibody assay, serological assays assessing immunoglobulin A or G, unclear)

Risk of bias variables

- Validity of the infertility definition (follows WHO definition that is failure to conceive after at least one year of regular unprotected intercourse, des not follow WHO definition, and unclear)
- Lack of exposure to antimicrobials for at least one week prior to collection of biological samples (ascertained, not ascertained, and unclear)
- Consistency in assay used for infection ascertainment (same assay used to test all participants, different assays used to test all participants, and unclear)
- Response rate (≥80%, <80%, and unclear)

Supplementary Table 2 Countries covered under the different World Health Organization regions[2]

World Health Organization region	Countries
African Region (AFRO)	Algeria, Angola, Benin, Botswana, Burkina Faso, Burundi, Cameroon, Cabo Verde, Central African Republic, Chad, Comoros, Congo, Côte d'Ivoire, Democratic Republic of the Congo, Equatorial Guinea, Eritrea, Ethiopia, Gabon, Gambia, Ghana, Guinea, Guinea-Bissau, Kenya, Lesotho, Liberia, Madagascar, Malawi, Mali, Mauritania, Mauritius, Mozambique, Namibia, Niger, Nigeria, Rwanda, Sao Tome and Principe, Senegal, Seychelles, Sierra Leone, South Africa, South Sudan, Swaziland, Togo, Uganda, United Republic of Tanzania, Zambia, Zimbabwe.
Region of the Americas (AMRO)	Antigua and Barbuda, Argentina, Bahamas, Barbados, Belize, Bolivia (Plurinational State of), Brazil, Canada, Chile, Colombia, Costa Rica, Cuba, Dominica, Dominican Republic, Ecuador, El Salvador, Grenada, Guatemala, Guyana, Haiti, Honduras, Jamaica, Mexico, Nicaragua, Panama, Paraguay, Peru, Saint Kitts and Nevis, Saint Lucia, Saint Vincent and the Grenadines, Suriname, Trinidad and Tobago, United States of America, Uruguay, Venezuela (Bolivarian Republic of).
Eastern Mediterranean Region (EMRO)	Afghanistan, Bahrain, Djibouti, Egypt, Iran (Islamic Republic of), Iraq, Jordan, Kuwait, Lebanon, Libya, Morocco, Oman, Pakistan, Qatar, Saudi Arabia, Somalia, Sudan, Syrian Arab Republic, Tunisia, United Arab Emirates, Yemen.
European Region (EURO)	Albania, Andorra, Armenia, Austria, Azerbaijan, Belarus, Belgium, Bosnia and Herzegovina, Bulgaria, Croatia, Cyprus, Czech Republic, Denmark, Estonia, Finland, France, Georgia, Germany, Greece, Hungary, Iceland, Ireland, Israel, Italy, Kazakhstan, Kyrgyzstan, Latvia, Lithuania, Luxembourg, Malta, Monaco, Montenegro, Netherlands, Norway, Poland, Portugal, Republic of Moldova, Romania, Russian Federation, San Marino, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland, Tajikistan, The former Yugoslav Republic of Macedonia, Turkey, Turkmenistan, Ukraine, United Kingdom, Uzbekistan.
South-East Asia Region (SEARO)	Bangladesh, Bhutan, Democratic People's Republic of Korea, India, Indonesia, Maldives, Myanmar, Nepal, Sri Lanka, Thailand, Timor-Leste.
Western Pacific Region (WPRO)	Australia, Brunei Darussalam, Cambodia, China, Cook Islands, Fiji, Japan, Kiribati, Lao People's Democratic Republic, Malaysia, Marshall Islands, Micronesia (Federated States of), Mongolia, Nauru, New Zealand, Niue, Palau, Papua New Guinea, Philippines, Republic of Korea, Samoa, Singapore, Solomon Islands, Tonga, Tuvalu, Vanuatu, Viet Nam.

Supplementary Table 3 Studies reporting *Neisseria gonorrhoeae* (NG) prevalence in infertile populations in the World Health Organization African Region (AFRO)*

Country	First author, year	Study site	Study period	Population characteristics	Infertility type	Specimen/Assay-biomarker	Tested (N)	NG prev (%)†
AFRO (20 studies)								
Cameroon	Merle, 1960 [3]	Infertility clinic	NR	Women-Majority TFI	Not specified	Unclear/Unclear	50	2.0
Cote d'Ivoire	Faye-Kette, 1995 [4]	Infertility clinic	NR	Men-Unclear diagnosis	Not specified	Urethral/Gram stain	58	26.0
Gabon	Collet, 1988 [5]	GYN clinic	1983-84	Women-No TFI	Majority secondary	Endocervical/Culture	38	0
Gabon	Collet, 1988 [5]	GYN clinic	1983-84	Women-TFI	Majority secondary	Endocervical/Culture	181	5.6
Gambia	Mabey, 1985 [6]	Infertility clinic	NR	Women-TFI	Majority secondary	Serum/ELISA-IgG	37	65
Gambia	Walraven, 2001 [7]	Community	1999	Women-Unclear diagnosis	Majority secondary	Endocervical/Culture	85	0
Kenya	Cohen, 2000 [8]	GYN clinic	1995-96	Women-TFI	Majority secondary	Endocervical/Culture	47	0
Nigeria	Arowojolu, 1998 [9]	Hospital	NR	Women-TFI	Not specified	Endocervical/Culture	39	46.2
Nigeria	Nsonwu-Anyanwu [10]	Infertility clinic	NR	Women-TFI	Not specified	Endocervical/Gram stain	40	0
Nigeria	Ogunbanjo, 1989 [11]	Infertility clinic	1975-84	Men-Male factor	Not specified	Semen/Culture	782	1.5
Nigeria	Okonofua, 1995 [12]	Hospital	1990-91	Women-Mixed diagnosis	Majority secondary	Endocervical & Vaginal/Culture	92	17.4
Nigeria	Okonofua, 1997 [13]	Community	1994-95	Women-Unclear diagnosis	Majority secondary	Serum/ELISA-IgG	60	26.7
Rwanda	Muvunyi, 2011 [14]	Infertility clinic	2007-10	Women-Unclear diagnosis	Not specified	Vaginal/NAAT	242	4.1
Senegal	Cisse, 1997 [15]	GYN clinic	1991-95	Women-Majority with TFI	Majority primary	Vaginal/Unclear	82	0
South Africa	Duba, 2017 [16]	Infertility clinic	NR	Women-Unclear diagnosis	Not specified	Vaginal/NAAT	51	0
South Africa	Hoosen, 1996 [17]	GYN clinic	NR	Women-Unclear diagnosis	Not specified	Endocervical/Culture	80	2.5
Tanzania	Mandara, 1980 [18]	GYN clinic	1978	Women-Unclear diagnosis	Not specified	Endocervical/Culture	64	4.7
Uganda	Arya, 1980 [19]	Community	1971-72	Women-Unclear diagnosis	Not specified	Endocervical/Culture	30	10.0
Zimbabwe	De Muylder, 1990 [20]	Hospital	1985-87	Women-TFI	Majority secondary	Serum/ELISA-IgG	135	51.5
Zimbabwe	De Muylder, 1990 [20]	Hospital	1985-87	Women-No TFI	Majority secondary	Serum/ELISA-IgG	92	25.0

ELISA: enzyme-linked immunoassay; GYN: gynecology clinic; IgG: immunoglobulin G; NAAT: nucleic acid amplification test; NR: not reported; Prev: prevalence; TFI: tubal factor infertility.
*The table is sorted by country and first author's name.
†The decimal places of the prevalence measures are as reported in the original report, but prevalence figures with more than one decimal places were rounded to one decimal place.

Supplementary Table 4 Studies reporting *Neisseria gonorrhoeae* (NG) prevalence in infertile populations in the World Health Organization Region of the Americas (AMRO)*

Country	First author, year	Study site	Study period	Population characteristics	Infertility type	Specimen/Assay-biomarker	Tested (N)	NG prev (%)†
AMRO (24 studies)								
Argentina	Giovannini, 1991 [21]	Infertility clinic	NR	Women-Unclear diagnosis	Not specified	Endocervical/Culture	46	0
Argentina	Giovannini, 1991 [21]	Infertility clinic	NR	Men-Unclear diagnosis	Not specified	Urethral/Culture	24	0
Argentina	Videla, 1994 [22]	Hospital	1989-91	Women-TFI	Majority secondary	Endocervical/Culture	32	3.1
Brazil	Fernandes, 2014 [23]	Hospital	2009-12	Women-Mixed diagnosis	Majority secondary	Urine/NAAT	340	0.6
Brazil	Pantoja, 2012 [24]	IVF clinic	2008-09	Women-TFI	Majority secondary	Endocervical/NAAT	63	0
Brazil	Pantoja, 2012 [24]	IVF clinic	2008-09	Women-No TFI	Majority secondary	Endocervical/NAAT	47	0
Brazil	Piscopo, 2018 [25]	Infertility clinic	NR	Women-TFI	Not specified	Endocervical/NAAT	83	0
Brazil	Piscopo, 2018 [25]	Infertility clinic	NR	Women-No TFI	Not specified	Endocervical/NAAT	73	0
Canada	Brunham, 1988 [26]	Hospital	1983-87	Women-Mixed diagnosis	Not specified	Endocervical/Culture	11	18.2
Canada	Domes, 2012 [27]	Infertility clinic	2003-10	Women-Unclear diagnosis	Not specified	Urine/NAAT	5,588	0.02
Chile	Costoya, 2012 [28]	Hospital	2008-10	Women-TFI	Not specified	Endocervical/NAAT	30	10.0
Mexico	Cravioto Mdel, 2003 [29]	Hospital	1992-93	Women-No TFI	Not specified	Serum/ELISA-IgA	115	7.8
Mexico	Cravioto Mdel, 2003 [29]	Hospital	1992-93	Women-TFI	Not specified	Serum/ELISA-IgA	71	9.8
Mexico	Cravioto Mdel, 2003 [29]	Hospital	1992-93	Men-Male factor	Not specified	Serum/ELISA-IgA	74	6.7
Mexico	Cravioto Mdel, 2003 [29]	Hospital	1992-93	Women-No TFI	Not specified	Serum/ELISA-IgG	115	10.4
Mexico	Cravioto Mdel, 2003 [29]	Hospital	1992-93	Women-TFI	Not specified	Serum/ELISA-IgG	71	5.6
Mexico	Cravioto Mdel, 2003 [29]	Hospital	1992-93	Men-Male factor	Not specified	Serum/ELISA-IgG	74	1.3
Mexico	Escobedo-Guerra, 2018 [30]	Infertility clinic	2016	Women-Unclear diagnosis	Not specified	Endocervical/NAAT	250	24.0
United States	Gump, 1983 [31]	Infertility clinic	1978-82	Women-Unclear diagnosis	Not specified	Endocervical/Culture	203	0
United States	Imudia, 2008 [32]	Hospital	2001-07	Women-Unclear diagnosis	Not specified	Endocervical/Unclear	236	0.4
United States	Kundsin, 1986 [33]	Infertility clinic	NR	Women-Unclear diagnosis	Majority primary	Unclear/Culture	99	0
United States	Martens, 1993 [34]	Infertility clinic	NR	Women-Mixed diagnosis	Majority primary	Endocervical/Culture	79	0
United States	Pavletic, 1999 [35]	Hospital	1983-1990	Women-Majority with TFI	Not specified	Unclear/Culture	19	53
United States	Witkin, 1994 [36]	IVF clinic	1992	Women-Mixed diagnosis	Not specified	Endocervical/Culture	216	0

ELISA: enzyme-linked immunoassay; IgG: immunoglobulin G; IVF: in-vitro fertilization; NAAT: nucleic acid amplification test; NR: not reported; Prev: prevalence; TFI: tubal factor infertility.
*The table is sorted by country and first author's name.
†The decimal places of the prevalence measures are as reported in the original report, but prevalence figures with more than one decimal places were rounded to one decimal place.

Supplementary Table 5 Studies reporting *Neisseria gonorrhoeae* (NG) prevalence in infertile populations in the World Health Organization Eastern Mediterranean Region (EMRO)*

Country	First author, year	Study site	Study period	Population characteristics	Infertility type	Specimen/Assay-biomarker	Tested (N)	NG prev (%)†
EMRO (11 studies)								
Egypt	Shaaban, 1994 [37]	Hospital	NR	Women-TFI	Not specified	Endocervical/Culture	150	0.7
Iran	Azizmohammadi, 2016 [38]	Hospital	2015	Women-Unclear diagnosis	Not specified	Vaginal/Culture	130	16.9
Iran	Golshani, 2006 [39]	Infertility clinic	NR	Men-Unclear diagnosis	Not specified	Semen/Culture	88	2.3
Iraq	Al-Janabi, 2014 [40]	Infertility clinic	NR	Men-Male factor	Not specified	Semen/Culture	583	1.0
Jordan	Abusarah, 2013 [41]	GYN Clinic	2011	Men-Male factor	Not specified	Urine/NAAT	81	0
Saudi Arabia	Bakir, 1989 [42]	Hospital	NR	Women-Unclear diagnosis	Not specified	Endocervical/Other‡	37	0
Sudan	Almroth, 2005 [43]	Infertility clinic	2003-04	Women-Mixed diagnosis	Primary	Serum/EIA-IgG	79	3
Tunisia	Gdoura, 2008 [44]	Infertility clinic	NR	Women-Unclear diagnosis	Not specified	Semen/Culture	116	0
Tunisia	Gdoura, 2008 [44]	Infertility clinic	NR	Women-Unclear diagnosis	Not specified	Semen/NAAT	116	0
Tunisia	Sellami, 2014 [45]	GYN Clinic	NR	Men-Partners in infertile couples	Not specified	Semen/Culture	85	0
Tunisia	Sellami, 2014 [45]	GYN Clinic	NR	Men-Partners in infertile couples	Not specified	Semen/NAAT	85	5.8

EIA: enzyme immunoassay; GYN: gynecology clinic; IgG: immunoglobulin G; NAAT: nucleic acid amplification test; NR: not reported; Prev: prevalence; TFI: tubal factor infertility.
*The table is sorted by country and first author's name.
†The decimal places of the prevalence measures are as reported in the original report, but prevalence figures with more than one decimal places were rounded to one decimal place.
‡Gonozyme immunoassay was used (antigen detection).

Supplementary Table 6 Studies reporting *Neisseria gonorrhoeae* (NG) prevalence in infertile populations in the World Health Organization European Region (EURO)*

Country	First author, year	Study site	Study period	Population characteristics	Infertility type	Specimen/Assay-biomarker	Tested (N)	NG prev (%)†
EURO (65 studies)								
Armenia	Oganesian, 1990 [46]	Infertility clinic	NR	Women-Unclear diagnosis	Not specified	Endocervical/Culture	200	21.5
Armenia	Oganesian, 1990 [46]	Infertility clinic	NR	Men-Unclear diagnosis	Not specified	Urethral/Culture	200	22.5
Armenia	Oganesian, 1990 [46]	Infertility clinic	NR	Women-Unclear diagnosis	Not specified	Serum/Unclear	200	31.0
Armenia	Oganesian, 1990 [46]	Infertility clinic	NR	Men-Unclear diagnosis	Not specified	Serum/Unclear	200	28.5
Austria	Dietrich, 2010 [47]	Hospital	2006-07	Women-Majority no TFI	Majority primary	Endocervical/NAAT	202	0
Croatia	Dolenec,1980 [48]	Infertility clinic	NR	Women-Unclear diagnosis	Not specified	Endocervical/Culture	61	1.6
Croatia	Plecko, 2014 [49]	Hospital	NR	Men-Male factor	Not specified	Urine/NAAT	145	0.7
Czech Republic	Drasnar, 1978 [50]	Infertility clinic	NR	Women-TFI	Not specified	Endocervical/Culture	92	52.2
Czech Republic	Klanica, 1978 [50]	GYN clinic	1976-77	Women-Unclear diagnosis	Not specified	Endocervical/Culture	63	33.3
Czech Republic	Klanica, 1978 [50]	GYN clinic	1976-77	Women-Unclear diagnosis	Not specified	Endocervical/Other‡	63	23.8
Denmark	Kjaergaard, 1997 [51]	Infertility clinic	1993-95	Men-Unclear diagnosis	Not specified	Semen/Culture	201	0
Denmark	Moller, 1984 [52]	GYN clinic	1981-82	Women-Majority no TFI	Majority primary	Endocervical/Culture	116	0
Finland	Koskimies, 1981 [53]	Infertility clinic	1978-79	Women-Cervical infertility	Majority primary	Endocervical/Culture	52	0
Finland	Miettinen, 1990 [54]	IVF clinic	1984-86	Women-Majority with TFI	Majority secondary	Serum/EIA-IgG	104	10.6
Finland	Punnonen, 1979 [55]	Infertility clinic	NR	Women-Mixed diagnosis	Majority primary	Unclear/Unclear	128	6.3
Finland	Puolakkainen, 1986 [56]	GYN clinic	1983	Women-Unclear diagnosis	Not specified	Endocervical/Culture	12	41.7
France	Alexandre, 1971 [57]	Hospital	NR	Men-Male factor	Majority primary	Unclear/Unclear	534	7.5
France	Diquelou, 1988 [58]	GYN clinic	1983-87	Men-Mixed diagnosis	Not specified	Urethral/Culture	80	0
France	Guillet-Rosso, 1987 [59]	IVF clinic	1983-84	Men-Unclear diagnosis	Not specified	Semen/Culture	519	0
France	Henry-Suchet, 1980 [60]	Hospital	1979	Women-TFI	Not specified	Unclear/Culture	17	0
France	Henry-Suchet, 1980 [60]	Hospital	1979	Women-TFI	Not specified	Unclear/Culture	29	0
France	Henry-Suchet, 1983 [61]	Hospital	1981-82	Women-TFI	Not specified	Vaginal/Unclear	16	0
France	Hermabessiere, 1977 [62]	Infertility clinic	NR	Men-Male factor	Not specified	Semen/Culture	53	1.9
France	Raiga, 1996 [63]	GYN clinic	1983-92	Women-Majority with TFI	Majority secondary	Endocervical/Unclear	24	4.2
Georgia	Durglishvili, 2013 [64]	Infertility clinic	NR	Men-Male factor	Primary	Urethral/Gram stain	54	29.6
Georgia	Tchiokadze & Galdava, 2015 [65]	Infertility clinic	NR	Men-Unclear diagnosis	Not specified	Serum/Unclear	496	3.4
Germany	Eggert-Kruse, 1987 [66]	Infertility clinic	NR	Women-Majority no TFI	Majority primary	Endocervical/Culture	233	0
Germany	Eggert-Kruse, 1987 [66]	Infertility clinic	NR	Men-Majority with male factor	Majority primary	Semen/Culture	230	0
Germany	Eggert-Kruse, 1995 [67]	Infertility clinic	NR	Women-Majority no TFI	Majority primary	Endocervical/Culture	163	0
Germany	Eggert-Kruse, 2015 [68]	Infertility clinic	NR	Men-Unclear diagnosis	Majority primary	Semen/Culture	128	0
Germany	Graspeuntner, 2018 [69]	Hospital	2012-14	Women-TFI	Not specified	Endocervical/NAAT	21	0
Germany	Graspeuntner, 2018 [69]	Hospital	2012-14	Women-No TFI	Not specified	Endocervical/NAAT	26	0
Germany	Hofmann, 1985 [70]	Infertility clinic	1983-84	Women-Unclear diagnosis	Not specified	Endocervical/Culture	121	0
Germany	Hofmann, 1985 [70]	Infertility clinic	1983-84	Women-Unclear diagnosis	Not specified	Endocervical/EIA	121	0
Germany	Kolmorgen, 1987 [71]	Infertility clinic	NR	Women-TFI (bilateral)	Not specified	Endocervical/Culture	88	21.6
Germany	Kolmorgen, 1987 [71]	Infertility clinic	NR	Women-TFI (unilateral)	Not specified	Endocervical/Culture	38	28.9
Ireland	Walsh, 2011 [72]	IVF clinic	2009-11	Women-Unclear diagnosis	Not specified	Urine/NAAT	225	0
Italy	Cevenini, 1982 [73]	GYN clinic	NR	Women-Idiopathic infertility	Not specified	Endocervical/Culture	30	0
Italy	Cevenini, 1982 [73]	GYN clinic	NR	Women-TFI	Not specified	Endocervical/Culture	40	0
Italy	Cevenini, 1982 [73]	GYN clinic	NR	Men-Partners in infertile couple	Not specified	Urethral/Culture	30	0
Italy	Lucisano, 1992 [74]	GYN clinic	NR	Women-Mixed diagnosis	Not specified	Endocervical/Unclear	83	0
Italy	Moreno, 2018 [75]	IVF clinic	NR	Women-Unclear diagnosis	Not specified	Vaginal/NAAT	95	0
Italy	Semmola, 1967 [76]	Infertility clinic	1952-66	Men-Mixed diagnosis	Not specified	Semen/Culture	270	1.1

Italy	Vicari, 1986 [77]	Infertility clinic	NR	Men-Idiopathic infertility	Majority primary	Semen/Culture	90	10.0
Italy	Vicari, 1991 [78]	Infertility clinic	1985-89	Men-Male factor	Primary	Semen/Culture	785	14.1
Netherlands	Peeters, 1988 [79]	IVF clinic	1986	Women-Majority with TFI	Not specified	Serum/ELISA-IgG	77	11.7
Netherlands	Tjiam, 1985 [80]	Infertility clinic	NR	Women-Majority with TFI	Majority secondary	Serum/ELISA-IgG	53	47.2
Norway	Tuveng, 1985 [81]	Hospital	1982-83	Women-Majority no TFI	Not specified	Endocervical/Culture	80	0
Poland	Mierzejewski, 1987 [82]	Infertility clinic	NR	Men-Male factor	Not specified	Semen/Culture	61	1.6
Poland	Szymanowski, 1997 [83]	Infertility clinic	1993-95	Women-Unclear diagnosis	Not specified	Endocervical/Culture	72	0
Poland	Tomusiak, 2013 [84]	Infertility clinic	NR	Women-Unclear diagnosis	Not specified	Endocervical/NAAT	101	0
Romania	Mares, 2009 [85]	Infertility clinic	2008-09	Women-Unclear diagnosis	Majority secondary	Endocervical/Culture	125	0.8
Romania	Miron, 2013 [86]	GYN clinic	2009-12	Women-Unclear diagnosis	Not specified	Vaginal/NAAT	176	0.6
Russia	Mezinova, 1963 [87]	Infertility clinic	NR	Women-Majority with TFI	Majority secondary	Unclear/Unclear	100	7.0
Russia	Molzhaninov, 1982 [88]	Infertility clinic	NR	Women-Mixed diagnosis	Majority primary	Unclear/Unclear	209	14.8
Russia	Tishchenko, 1988 [89]	Infertility clinic	NR	Men-Male factor	Not specified	Urethral/Unclear	240	0
Russia	Uzlova, 2000 [90]	Infertility clinic	NR	Women-TFI	Majority primary	Endocervical/Culture	363	0.3
Spain	Rodriguez, 2003 [91]	Infertility clinic	NR	Women-Cervical factor	Majority primary	Endocervical/Culture	281	0.3
Sweden	Svensson, 1983 [92]	GYN clinic	1975-79	Women-Mixed diagnosis	Not specified	Endocervical/Culture	38	15.8
Turkey	Akgul, 2018 [93]	Infertility clinic	NR	Men-Male factor	Not specified	Urine and/or Semen/NAAT	25	0
Turkey	Akgul, 2018 [93]	Infertility clinic	NR	Men-Unclear diagnosis	Not specified	Urine and/or Semen/NAAT	25	0
Turkey	Eksi, 2011 [94]	GYN clinic	2008-09	Women-Unclear diagnosis	Not specified	Endocervical/NAAT	98	1.0
Turkey	Eldivan, 2016 [95]	IVF clinic	2009	Women-Mixed diagnosis	Not specified	Endocervical/Culture	60	0
United Kingdom	Robertson, 1987 [96]	Infertility clinic	NR	Women-TFI	Majority primary	Serum/ELISA-IgG	48	2
United Kingdom	Robertson, 1987 [96]	Infertility clinic	NR	Women-No TFI	Majority primary	Serum/ELISA-IgG	77	3

ELISA: enzyme-linked immunoassay; GYN: gynecology clinic; IgG: immunoglobulin G; IVF: in-vitro fertilization; NAAT: nucleic acid amplification test; NR: not reported; Prev: prevalence; TFI: tubal factor infertility.

*The table is sorted by country and first author's name.

†The decimal places of the prevalence measures are as reported in the original report, but prevalence figures with more than one decimal places were rounded to one decimal place.

‡Fluorescent antibody test was used.

Supplementary Table 7 Studies reporting *Neisseria gonorrhoeae* (NG) prevalence in infertile populations in the World Health Organization South-East Asia Region (SEARO)*

Country	First author, year	Study site	Study period	Population characteristics	Infertility type	Specimen/Assay-biomarker	Tested (N)	NG prev (%)†
SEARO (12 studies)								
India	Adamson, 2011 [97]	Hospital	2005-06	Women-Unclear diagnosis	Primary	Vaginal/Culture	113	0
India	Brabin, 1998 [98]	Hospital	1993-95	Women-Unclear diagnosis	Majority primary	Endocervical/Culture	295	0
India	Dhawan, 2014 [99]	Infertility clinic	2010-12	Women-Majority with TFI	Not specified	Endocervical/Culture	250	0.4
India	Dutta, 2008 [100]	GYN clinic	2005-06	Women-Unclear diagnosis	Majority secondary	Endocervical/ Culture	72	0
India	Gupta, 2009 [101]	GYN clinic	NR	Women-Unclear diagnosis	Not specified	Endocervical/ Culture	76‡	0
India	Jain, 1994 [102]	Hospital	NR	Women-Majority with TFI	Not specified	Unclear/Unclear	57	0
India	Reddy, 2004 [103]	Hospital	NR	Women-Majority with TFI	Majority primary	Endocervical/Culture	37	0
India	Tyagi & Singh, 1998 [104]	Hospital	NR	Women-TFI	Not specified	Serum/ELISA-IgG	81	4.9
Thailand	Swasdio, 1996 [105]	Hospital	1990-92	Women-TFI	Primary	Serum/ELISA-IgG	55	29.1
Thailand	Swasdio, 1996 [105]	Hospital	1990-92	Women-No TFI	Primary	Serum/ELISA-IgG	58	5.2
Thailand	Swasdio, 1996 [105]	Hospital	1990-92	Men-Partners of women with TFI	Not specified	Serum/ELISA-IgG	55	36.4
Thailand	Swasdio, 1996 [105]	Hospital	1990-92	Men-Partners of women with no TFI	Not specified	Serum/ELISA-IgG	58	8.6

ELISA: enzyme-linked immunoassay; GYN: gynecology clinic; IgG: immunoglobulin G; NR: not reported; Prev: prevalence; TFI: tubal factor infertility.
*The table is sorted by country and first author's name.
†The decimal places of the prevalence measures are as reported in the original report, but prevalence figures with more than one decimal places were rounded to one decimal place.
‡Mixed sample of infertile women who failed to achieve conception after 18 months of regular unprotected intercourse and women with pelvic inflammatory disease.

Supplementary Table 8 Studies reporting *Neisseria gonorrhoeae* (NG) prevalence in infertile populations in the World Health Organization Western Pacific Region (WPRO) and multicentre studies*

Country	First author, year	Study site	Study period	Population characteristics	Infertility type	Specimen/Assay-biomarker	Tested (N)	NG prev (%)†
WPRO (11 studies)								
Australia	Debattista, 2004 [106]	IVF clinic	NR	Women-Mixed diagnosis	Not specified	Endocervical/NAAT	44	0
Australia	Kildea, 2000 [107]	Community	1996	Women-Idiopathic infertility	Primary	Unclear/Culture	33‡	39
Australia	Kildea, 2000 [107]	Community	1996	Women-Idiopathic infertility	Secondary	Unclear/Culture	62‡	47
China	Han, 2016 [108]	Infertility clinic	2011-13	Men-Male factor	Not specified	Semen/Unclear	133	7.5
China	Li, 2017 [109]	Infertility clinic	2015	Women-Unclear diagnosis	Not specified	Vaginal/NAAT	133	0
China	Qing, 2017 [110]	Infertility clinic	2016	Men-Male factor	Primary	Urine/NAAT	2,607	0.2
China	Tan, 2007 [111]	Infertility clinic	NR	Men-Male factor	Not specified	Semen/Culture	41	4.9
China	Zhai, 2009 [112]	Hospital	2007-08	Women-Unclear diagnosis	Not specified	Endocervical/NAAT	76	32.9
Japan	Kobayashi, 2006 [113]	Hospital	1993-04	Women-TFI	Not specified	Endocervical/Culture	68	0
Korea	Bae, 2008 [114]	Hospital	2005-07	Women-TFI	Not specified	Unclear/Unclear	12	0
Malaysia	Hazlina, 2005 [115]	Infertility clinic	2002-03	Women-majority no TFI	Majority primary	Endocervical/Culture	150	0
Multi-center (4 studies)								
Thailand, Slovenia, Hungary	WHO, 1995 [116]	Infertility clinic	NR	Infertile women-Bilateral tubal occlusion	Majority primary	Endocervical/Culture	78	3
Thailand, Slovenia, Hungary	WHO, 1995 [116]	Infertility clinic	NR	Infertile women-Unilateral tubal occlusion and other infertility types	Majority primary	Endocervical/Culture	155	0
Thailand, Slovenia, Hungary	WHO, 1995 [116]	Infertility clinic	NR	Infertile women-Bilateral tubal occlusion	Majority primary	Serum/ELISA-IgG	45	62.2
Thailand, Slovenia, Hungary	WHO, 1995 [116]	Infertility clinic	NR	Infertile women-no TFI	Majority primary	Serum/ELISA-IgG	96	37.5

ELISA: enzyme-linked immunoassay; IgG: immunoglobulin G; IVF: in-vitro fertilization; NAAT: nucleic acid amplification test; NR: not reported; Prev: prevalence; TFI: tubal factor infertility.
*The table is sorted by country and first author's name.
†The decimal places of the prevalence measures are as reported in the original report, but prevalence figures with more than one decimal places were rounded to one decimal place.
‡Includes 11 women with resolved infertility.

Supplementary Table 9 Summary of precision and risk of bias assessments for *Neisseria gonorrhoeae* prevalence studies identified through the systematic review

Quality assessment	Studies identified through the systematic review	
	N	%
Precision		
Higher precision[*]	74	50.3
Lower precision	73	49.7
Risk of bias assessment		
Infertility definition		
Low risk of bias[†]	51	34.7
High risk of bias	2	1.3
Unclear[‡]	94	64.0
Lack of exposure to antimicrobials		
Low risk of bias[§]	21	14.3
High risk of bias	9	6.1
Unclear[‡]	117	79.6
Consistency in the assay used for infection ascertainment		
Low risk of bias[£]	142	96.6
High risk of bias	0	0
Unclear[‡]	5	3.4
Response rate		
Low risk of bias[ł]	1	0.7
High risk of bias	3	2.0
Unclear[‡]	143	97.3
Summary of risk of bias assessment for gonorrhea prevalence measures		
Low risk of bias		
In at least one quality domain	145	98.6
In at least two quality domains	61	41.5
In at least three quality domains	9	6.1
In all four quality domains	0	0
High risk of bias		
In at least one quality domain	13	8.8
In at least two quality domains	1	0.7
In at least three quality domains	0	0
In all four quality domains	0	0
Unclear risk of bias		
In at least one quality domain	146	99.3
In at least two quality domains	133	90.5
In at least three quality domains	78	53.1
In all four quality domains	2	1.4
Total studies	**147**	**100**

[*]Studies including ≥100 participants were considered of higher precision.
[†]Infertility definition follows the World Health Organization infertility definition, that is failure to conceive after at least one year of regular unprotected intercourse.
[‡]Reports with missing information for any of the domains were classified as having 'unclear' risk of bias for that specific domain.
[§]Lack of exposure to antimicrobials for at least one week prior to collection of biological samples.
[£]Same assay used for gonorrhea ascertainment for all participants.
[ł]Response rate ≥80%.

Supplementary Table 10 Precision and risk of bias assessments for *Neisseria gonorrhoeae* (NG) prevalence studies identified through the systematic review

Country	Short citation	Prev (%)	Precision*	Risk of bias domains			
				Infertility definition†,¦	Lack of exposure to antimicrobials‡,¦	Consistency in assay used for NG ascertainment§,¦	Response rate£,¦
AFRO							
Cameroon	Merle, 1960 [3]	2.0	Lower	Unclear	Unclear	Low ROB	Unclear
Cote d'Ivoire	Faye-Kette, 1995 [4]	26.0	Lower	Unclear	Unclear	Low ROB	Unclear
Gabon	Collet, 1988 [5]	0	Higher	Unclear	Unclear	Low ROB	Unclear
Gabon	Collet, 1988 [5]	5.6	Higher	Unclear	Unclear	Low ROB	Unclear
Gambia	Mabey, 1985 [6]	65	Lower	Low ROB	Unclear	Low ROB	Unclear
Gambia	Walraven, 2001 [7]	0	Lower	Low ROB	Unclear	Low ROB	<80%
Kenya	Cohen, 2000 [8]	0	Lower	Low ROB	High ROB	Low ROB	Unclear
Nigeria	Arowojolu, 1998 [9]	46.2	Lower	Unclear	Unclear	Low ROB	Unclear
Nigeria	Nsonwu-Anyanwu [10]	0	Lower	Low ROB	Unclear	Low ROB	Unclear
Nigeria	Ogunbanjo, 1989 [11]	1.5	Higher	Unclear	High ROB	Low ROB	Unclear
Nigeria	Okonofua, 1995 [12]	17.4	Lower	Low ROB	Unclear	Low ROB	Unclear
Nigeria	Okonofua, 1997 [13]	26.7	Lower	Low ROB	High ROB	Low ROB	<80%
Rwanda	Muvunyi, 2011 [14]	4.1	Higher	Unclear	Unclear	Low ROB	Unclear
Senegal	Cisse, 1997 [15]	0	Lower	Low ROB	Unclear	Low ROB	Unclear
South Africa	Duba, 2017 [16]	0	Lower	Unclear	Unclear	Low ROB	Unclear
South Africa	Hoosen, 1996 [17]	2.5	Lower	Unclear	Unclear	Low ROB	Unclear
Tanzania	Mandara, 1980 [18]	4.7	Lower	Unclear	Unclear	Low ROB	Unclear
Uganda	Arya, 1980 [19]	10.0	Lower	Unclear	Unclear	Low ROB	>=80%
Zimbabwe	De Muylder, 1990 [20]	51.5	Higher	Low ROB	Unclear	Low ROB	Unclear
Zimbabwe	De Muylder, 1990 [20]	25.0	Higher	Low ROB	Unclear	Low ROB	Unclear
AMRO							
Argentina	Giovannini, 1991 [21]	0	Lower	Unclear	High ROB	Low ROB	Unclear
Argentina	Giovannini, 1991 [21]	0	Lower	Unclear	High ROB	Low ROB	Unclear
Argentina	Videla, 1994 [22]	3.1	Lower	Unclear	Unclear	Low ROB	Unclear
Brazil	Fernandes, 2014 [23]	0.6	Higher	Low ROB	Unclear	Low ROB	Unclear
Brazil	Pantoja, 2012 [24]	0	Higher	Unclear	Unclear	Low ROB	Unclear
Brazil	Pantoja, 2012 [24]	0	Higher	Unclear	Unclear	Low ROB	Unclear
Brazil	Piscopo, 2018 [25]	0	Higher	Unclear	Unclear	Low ROB	Unclear
Brazil	Piscopo, 2018 [25]	0	Higher	Unclear	Unclear	Low ROB	Unclear
Canada	Brunham, 1988 [26]	18.2	Lower	Unclear	Low ROB	Low ROB	Unclear
Canada	Domes, 2012 [27]	0.02	Higher	Unclear	Unclear	Low ROB	Unclear
Chile	Costoya, 2012 [28]	10.0	Lower	Unclear	Low ROB	Unclear	Unclear
Mexico	Cravioto Mdel, 2003 [29]	7.8	Higher	Unclear	Unclear	Low ROB	Unclear
Mexico	Cravioto Mdel, 2003 [29]	9.8	Higher	Unclear	Unclear	Low ROB	Unclear
Mexico	Cravioto Mdel, 2003 [29]	6.7	Lower	Unclear	Unclear	Low ROB	Unclear
Mexico	Cravioto Mdel, 2003 [29]	10.4	Higher	Unclear	Unclear	Low ROB	Unclear
Mexico	Cravioto Mdel, 2003 [29]	5.6	Higher	Unclear	Unclear	Low ROB	Unclear
Mexico	Cravioto Mdel, 2003 [29]	1.3	Lower	Unclear	Unclear	Low ROB	Unclear
Mexico	Escobedo-Guerra, 2018 [30]	24.0	Higher	Unclear	Unclear	Low ROB	Unclear
United States	Gump, 1983 [31]	0	Higher	Low ROB	High ROB	Low ROB	Unclear
United States	Imudia, 2008 [32]	0.4	Higher	Unclear	Unclear	Low ROB	Unclear
United States	Kundsin, 1986 [33]	0	Lower	Unclear	Unclear	Low ROB	Unclear

United States	Martens, 1993 [34]	0	Lower	Low ROB	Unclear	Low ROB	Unclear
United States	Pavletic, 1999 [35]	53	Lower	Low ROB	Low ROB	Low ROB	Unclear
United States	Witkin, 1994 [36]	0	Higher	Unclear	Unclear	Low ROB	Unclear
EMRO							
Egypt	Shaaban, 1994 [37]	0.7	Higher	Unclear	Unclear	Low ROB	Unclear
Iran	Azizmohammadi, 2016 [38]	16.9	Higher	Low ROB	Unclear	Low ROB	Unclear
Iran	Golshani, 2006 [39]	2.3	Lower	Unclear	Low ROB	Low ROB	Unclear
Iraq	Al-Janabi, 2014 [40]	1.0	Higher	Unclear	Unclear	Low ROB	Unclear
Jordan	Abusarah, 2013 [41]	0	Lower	High ROB	Low ROB	Low ROB	Unclear
Saudi Arabia	Bakir, 1989 [42]	0	Lower	Unclear	Unclear	Low ROB	Unclear
Sudan	Almroth, 2005 [43]	3	Lower	Low ROB	Unclear	Low ROB	Unclear
Tunisia	Gdoura, 2008 [44]	0	Higher	Low ROB	Unclear	Low ROB	Unclear
Tunisia	Gdoura, 2008 [44]	0	Higher	Low ROB	Unclear	Low ROB	Unclear
Tunisia	Sellami, 2014 [45]	0	Lower	Low ROB	Unclear	Low ROB	Unclear
Tunisia	Sellami, 2014 [45]	5.8	Lower	Low ROB	Unclear	Low ROB	Unclear
EURO							
Armenia	Oganesian, 1990 [46]	21.5	Higher	Unclear	Unclear	Low ROB	Unclear
Armenia	Oganesian, 1990 [46]	22.5	Higher	Unclear	Unclear	Low ROB	Unclear
Armenia	Oganesian, 1990 [46]	31.0	Higher	Unclear	Unclear	Low ROB	Unclear
Armenia	Oganesian, 1990 [46]	28.5	Higher	Unclear	Unclear	Low ROB	Unclear
Austria	Dietrich, 2010 [47]	0	Higher	High ROB	Unclear	Low ROB	Unclear
Croatia	Dolenec,1980 [48]	1.6	Lower	Unclear	Unclear	Low ROB	Unclear
Croatia	Plecko, 2014 [49]	0.7	Higher	Unclear	Unclear	Low ROB	Unclear
Czech Republic	Drasnar, 1978 [50]	52.2	Lower	Unclear	Unclear	Low ROB	Unclear
Czech Republic	Klanica, 1978 [50]	33.3	Lower	Unclear	Unclear	Low ROB	Unclear
Czech Republic	Klanica, 1978 [50]	23.8	Lower	Unclear	Unclear	Low ROB	Unclear
Denmark	Kjaergaard, 1997 [51]	0	Higher	Unclear	Low ROB	Low ROB	Unclear
Denmark	Moller, 1984 [52]	0	Higher	Unclear	Low ROB	Low ROB	Unclear
Finland	Koskimies, 1981 [53]	0	Lower	Low ROB	Unclear	Low ROB	Unclear
Finland	Miettinen, 1990 [54]	10.6	Higher	Low ROB	Unclear	Low ROB	Unclear
Finland	Punnonen, 1979 [55]	6.3	Higher	Unclear	Unclear	Low ROB	Unclear
Finland	Puolakkainen, 1986 [56]	41.7	Lower	Unclear	Unclear	Low ROB	<80%
France	Alexandre, 1971 [57]	7.5	Higher	Low ROB	Unclear	Unclear	Unclear
France	Diquelou, 1988 [58]	0	Lower	Low ROB	Unclear	Low ROB	Unclear
France	Guillet-Rosso, 1987 [59]	0	Higher	Unclear	Unclear	Low ROB	Unclear
France	Henry-Suchet, 1980 [60]	0	Lower	Unclear	High ROB	Low ROB	Unclear
France	Henry-Suchet, 1980 [60]	0	Lower	Unclear	High ROB	Low ROB	Unclear
France	Henry-Suchet, 1983 [61]	0	Lower	Unclear	Unclear	Unclear	Unclear
France	Hermabessiere, 1977 [62]	1.9	Lower	Unclear	Unclear	Low ROB	Unclear
France	Raiga, 1996 [63]	4.2	Lower	Unclear	Unclear	Low ROB	Unclear
Georgia	Durglishvili, 2013 [64]	29.6	Lower	Low ROB	Unclear	Low ROB	Unclear
Georgia	Tchiokadze & Galdava, 2015 [65]	3.4	Higher	Low ROB	Unclear	Low ROB	Unclear
Germany	Eggert-Kruse, 1987 [66]	0	Higher	Low ROB	Low ROB	Low ROB	Unclear
Germany	Eggert-Kruse, 1987 [66]	0	Higher	Low ROB	Low ROB	Low ROB	Unclear
Germany	Eggert-Kruse, 1995 [67]	0	Higher	Low ROB	Unclear	Low ROB	Unclear
Germany	Eggert-Kruse, 2015 [68]	0	Higher	Low ROB	Low ROB	Low ROB	Unclear
Germany	Graspeuntner, 2018 [69]	0	Lower	Unclear	Unclear	Low ROB	Unclear
Germany	Graspeuntner, 2018 [69]	0	Lower	Unclear	Unclear	Low ROB	Unclear
Germany	Hofmann, 1985 [70]	0	Higher	Unclear	Unclear	Low ROB	Unclear
Germany	Hofmann, 1985 [70]	0	Higher	Unclear	Unclear	Low ROB	Unclear

Germany	Kolmorgen, 1987 [71]	21.6	Higher	Unclear	Unclear	Low ROB	Unclear
Germany	Kolmorgen, 1987 [71]	28.9	Higher	Unclear	Unclear	Low ROB	Unclear
Ireland	Walsh, 2011 [72]	0	Higher	Unclear	Unclear	Low ROB	Unclear
Italy	Cevenini, 1982 [73]	0	Lower	Unclear	Unclear	Low ROB	Unclear
Italy	Cevenini, 1982 [73]	0	Lower	Unclear	Unclear	Low ROB	Unclear
Italy	Cevenini, 1982 [73]	0	Lower	Unclear	Unclear	Low ROB	Unclear
Italy	Lucisano, 1992 [74]	0	Lower	Low ROB	Low ROB	Low ROB	Unclear
Italy	Moreno, 2018 [75]	0	Lower	Unclear	Unclear	Low ROB	Unclear
Italy	Semmola, 1967 [76]	1.1	Higher	Low ROB	Unclear	Low ROB	Unclear
Italy	Vicari, 1986 [77]	10.0	Lower	Unclear	Low ROB	Low ROB	Unclear
Italy	Vicari, 1991 [78]	14.1	Higher	Low ROB	Unclear	Low ROB	Unclear
Netherlands	Peeters, 1988 [79]	11.7	Lower	Unclear	Unclear	Low ROB	Unclear
Netherlands	Tjiam, 1985 [80]	47.2	Lower	Low ROB	Unclear	Low ROB	Unclear
Norway	Tuveng, 1985 [81]	0	Lower	Unclear	Low ROB	Low ROB	Unclear
Poland	Mierzejewski, 1987 [82]	1.6	Lower	Unclear	Unclear	Low ROB	Unclear
Poland	Szymanowski, 1997 [83]	0	Lower	Low ROB	Unclear	Low ROB	Unclear
Poland	Tomusiak, 2013 [84]	0	Higher	Low ROB	Low ROB	Low ROB	Unclear
Romania	Mares, 2009 [85]	0.8	Higher	Low ROB	Low ROB	Low ROB	Unclear
Romania	Miron, 2013 [86]	0.6	Higher	Low ROB	Low ROB	Low ROB	Unclear
Russia	Mezinova, 1963 [87]	7.0	Higher	Unclear	Unclear	Low ROB	Unclear
Russia	Molzhaninov, 1982 [88]	14.8	Higher	Unclear	Unclear	Low ROB	Unclear
Russia	Tishchenko, 1988 [89]	0	Higher	Unclear	Unclear	Low ROB	Unclear
Russia	Uzlova, 2000 [90]	0.3	Higher	Low ROB	Unclear	Low ROB	Unclear
Spain	Rodriguez, 2003 [91]	0.3	Higher	Low ROB	Unclear	Low ROB	Unclear
Sweden	Svensson, 1983 [92]	15.8	Lower	Low ROB	Unclear	Low ROB	Unclear
Turkey	Akgul, 2018 [93]	0	Lower	Unclear	Low ROB	Low ROB	Unclear
Turkey	Akgul, 2018 [93]	0	Lower	Unclear	Low ROB	Low ROB	Unclear
Turkey	Eksi, 2011 [94]	1.0	Lower	Low ROB	Low ROB	Low ROB	Unclear
Turkey	Eldivan, 2016 [95]	0	Lower	Unclear	Low ROB	Unclear	Unclear
United Kingdom	Robertson, 1987 [96]	2	Higher	Unclear	Unclear	Low ROB	Unclear
United Kingdom	Robertson, 1987 [96]	3	Higher	Unclear	Unclear	Low ROB	Unclear
SEARO							
India	Adamson, 2011 [97]	0	Higher	Low ROB	Unclear	Low ROB	Unclear
India	Brabin, 1998 [98]	0	Higher	Unclear	High ROB	Low ROB	Unclear
India	Dhawan, 2014 [99]	0.4	Higher	Unclear	Unclear	Low ROB	Unclear
India	Dutta, 2008 [100]	0	Lower	Unclear	Unclear	Low ROB	Unclear
India	Gupta, 2009 [101]	0	Lower	Low ROB	Unclear	Low ROB	Unclear
India	Jain, 1994 [102]	0	Lower	Unclear	Unclear	Low ROB	Unclear
India	Reddy, 2004 [103]	0	Lower	Unclear	Low ROB	Low ROB	Unclear
India	Tyagi & Singh, 1998 [104]	4.9	Lower	Unclear	Unclear	Low ROB	Unclear
Thailand	Swasdio, 1996 [105]	29.1	Higher	Unclear	Unclear	Low ROB	Unclear
Thailand	Swasdio, 1996 [105]	5.2	Higher	Unclear	Unclear	Low ROB	Unclear
Thailand	Swasdio, 1996 [105]	36.4	Higher	Unclear	Unclear	Low ROB	Unclear
Thailand	Swasdio, 1996 [105]	8.6	Higher	Unclear	Unclear	Low ROB	Unclear
WPRO							
Australia	Debattista, 2004 [106]	0	Lower	Low ROB	Unclear	Low ROB	Unclear
Australia	Kildea, 2000 [107]	39	Lower	Low ROB	Unclear	Low ROB	Unclear
Australia	Kildea, 2000 [107]	47	Lower	Low ROB	Unclear	Low ROB	Unclear
China	Han, 2016 [108]	7.5	Higher	Unclear	Unclear	Low ROB	Unclear
China	Li, 2017 [109]	0	Higher	Unclear	Unclear	Low ROB	Unclear

China	Qing, 2017 [110]	0.2	Higher	Low ROB	Unclear	Low ROB	Unclear
China	Tan, 2007 [111]	4.9	Lower	Unclear	Unclear	Low ROB	Unclear
China	Zhai, 2009 [112]	32.9	Lower	Unclear	Unclear	Low ROB	Unclear
Japan	Kobayashi, 2006 [113]	0	Lower	Unclear	Unclear	Low ROB	Unclear
Korea	Bae, 2008 [114]	0	Lower	Unclear	Unclear	Unclear	Unclear
Malaysia	Hazlina, 2005 [115]	0	Higher	Unclear	Unclear	Low ROB	Unclear
Multi-center							
Thailand, Slovenia, Hungary	WHO, 1995 [116]	3	Higher	Low ROB	Unclear	Low ROB	Unclear
Thailand, Slovenia, Hungary	WHO, 1995 [116]	0	Higher	Low ROB	Unclear	Low ROB	Unclear
Thailand, Slovenia, Hungary	WHO, 1995 [116]	62.2	Higher	Low ROB	Unclear	Low ROB	Unclear
Thailand, Slovenia, Hungary	WHO, 1995 [116]	37.5	Higher	Low ROB	Unclear	Low ROB	Unclear

Prev: prevalence.

[*]Studies including ≥100 participants were considered of higher precision.

[†]Infertility definition follows the World Health Organization infertility definition, that is failure to conceive after at least one year of regular unprotected intercourse.

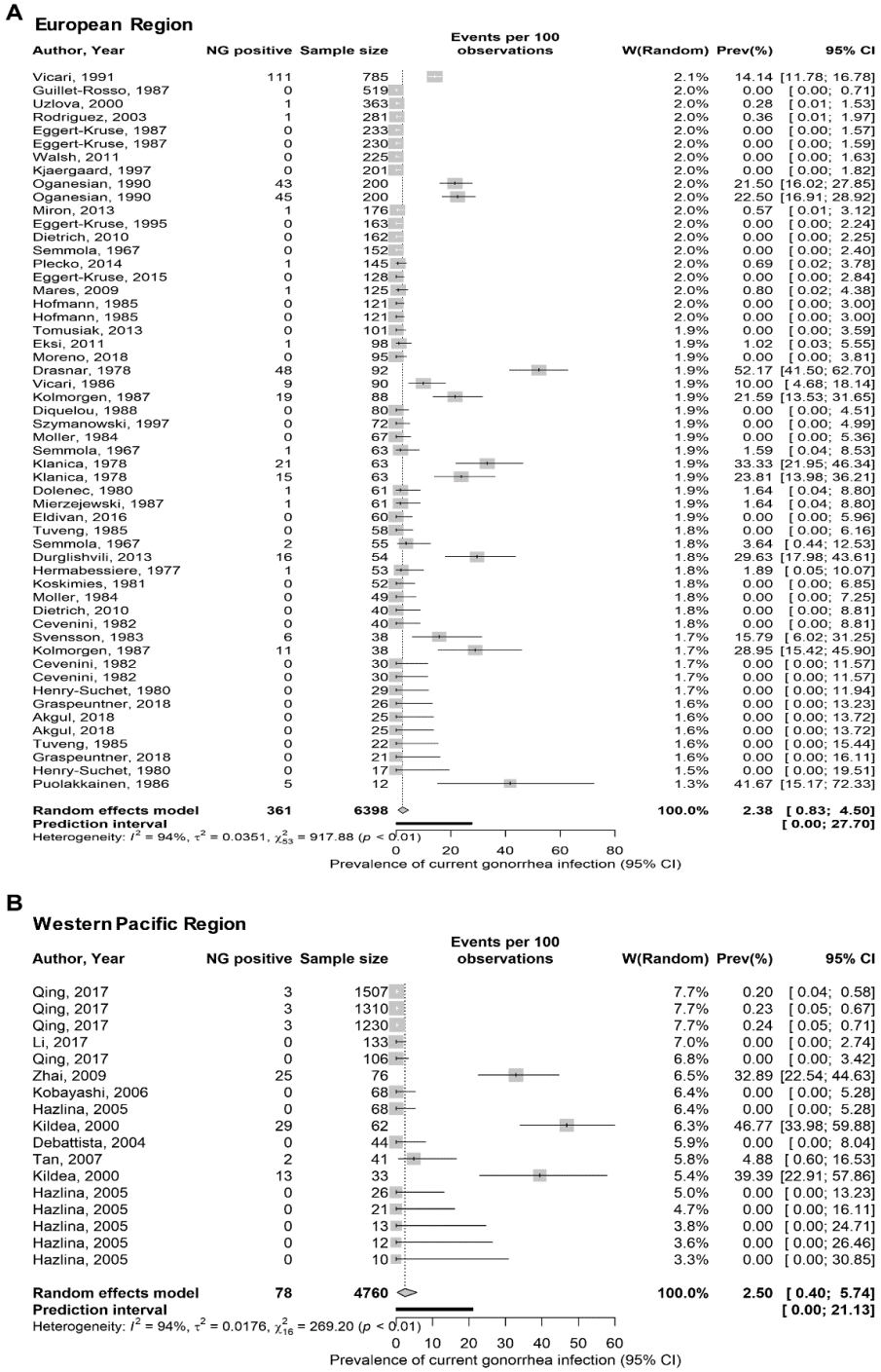
[‡]Lack of exposure to antimicrobials for at least one week prior to collection of biological samples.

[§]Same assay used for gonorrhea ascertainment for all participants.

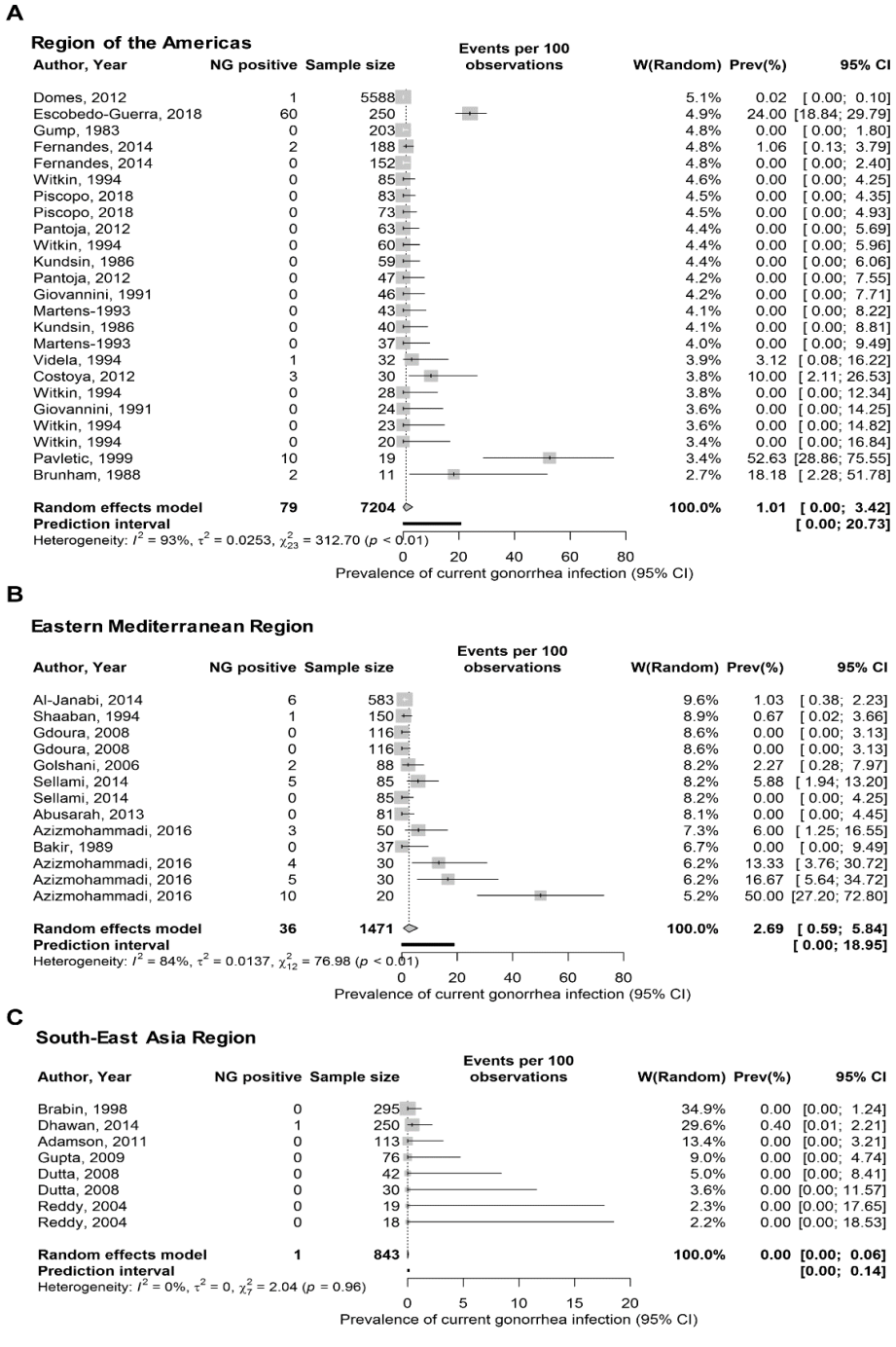
[£]Response rate ≥80%.

[ł]Reports with missing information for any of the domains were classified as having 'unclear' risk of bias for that specific domain.

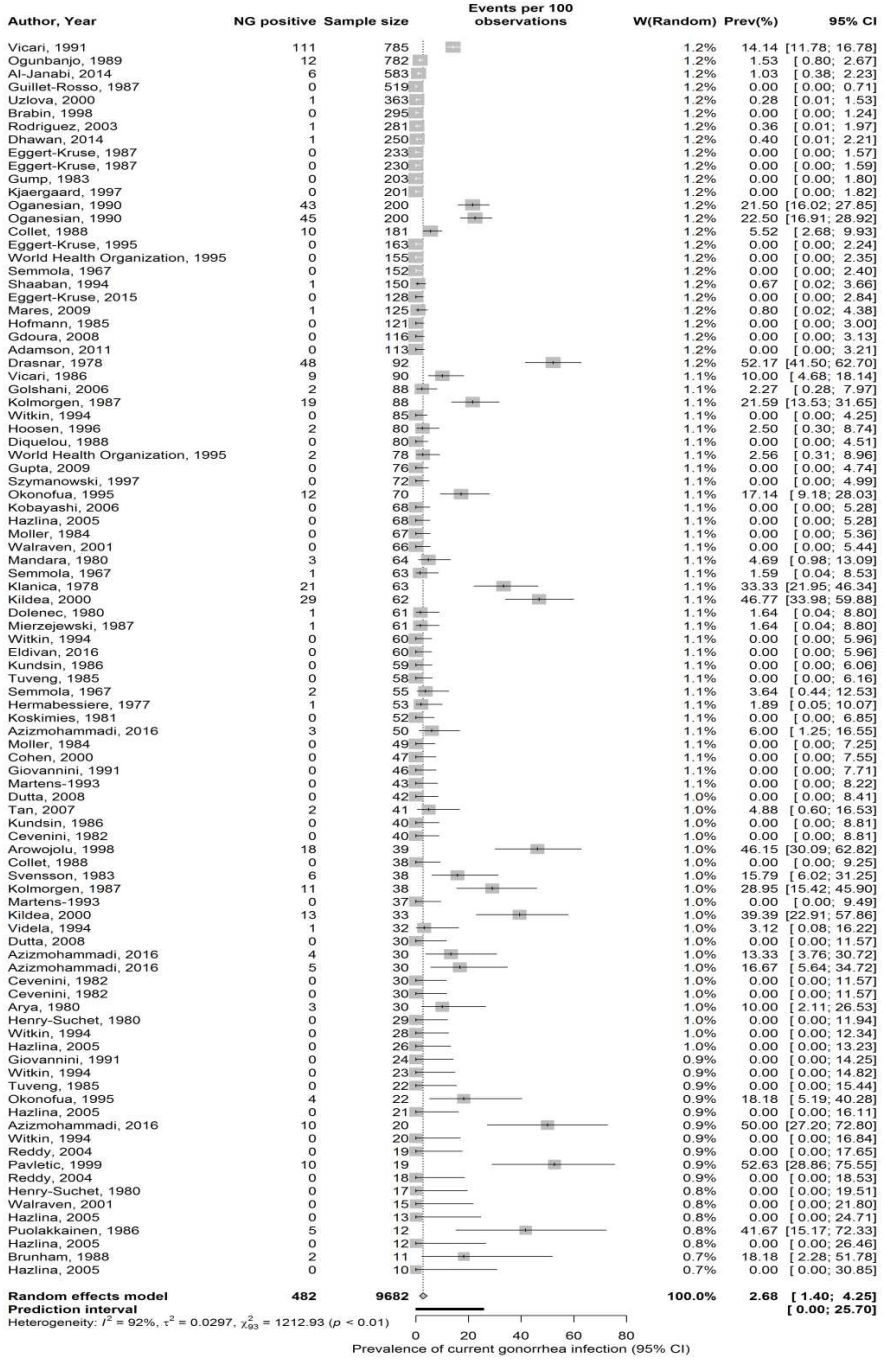
Supplementary Figure 1 Forest plots showing the results of meta-analysis on studies reporting the prevalence of current *Neisseria gonorrhoeae* (NG) infection in infertile populations for the World Health Organization A) European Region, and B) Western Pacific Region



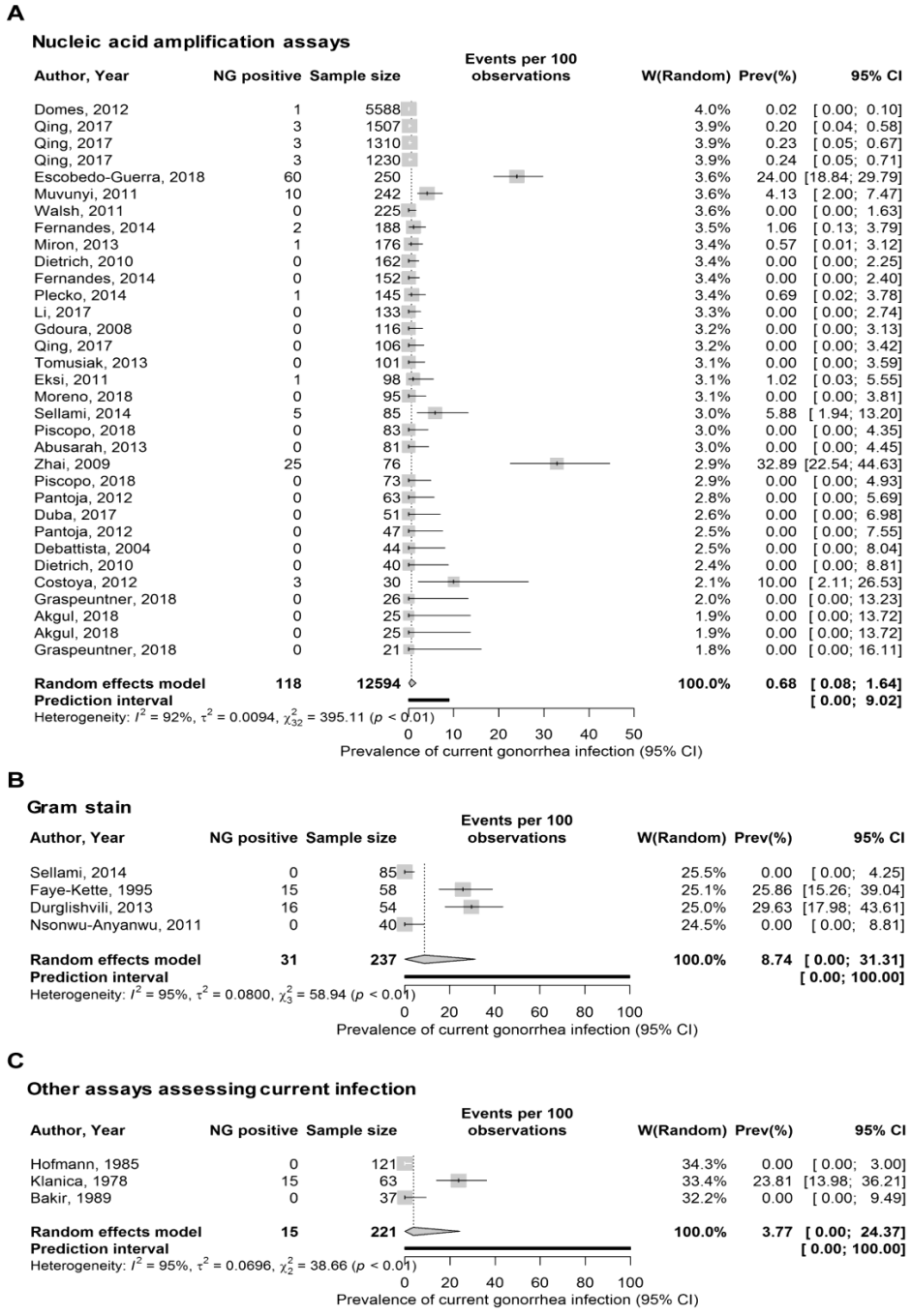
Supplementary Figure 2 Forest plots showing the results of meta-analysis on studies reporting the prevalence of current *Neisseria gonorrhoeae* (NG) infection in infertile populations for the World Health Organization A) Region of the Americas, B) Eastern Mediterranean Region, and C) South-East Asia Region



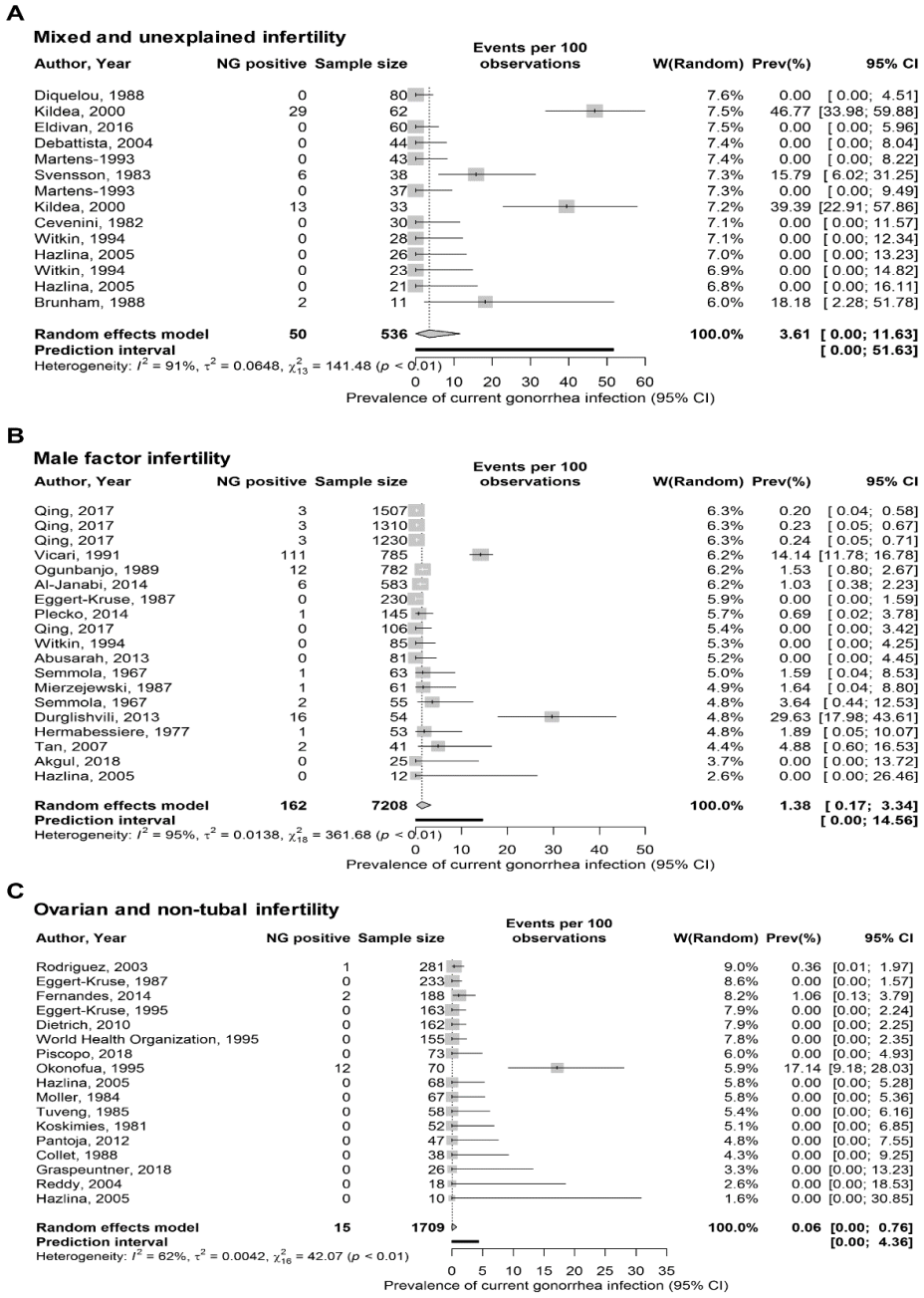
Supplementary Figure 3 Forest plot showing the results of meta-analysis on studies reporting the prevalence of current *Neisseria gonorrhoeae* (NG) infection in infertile populations as assessed using culture



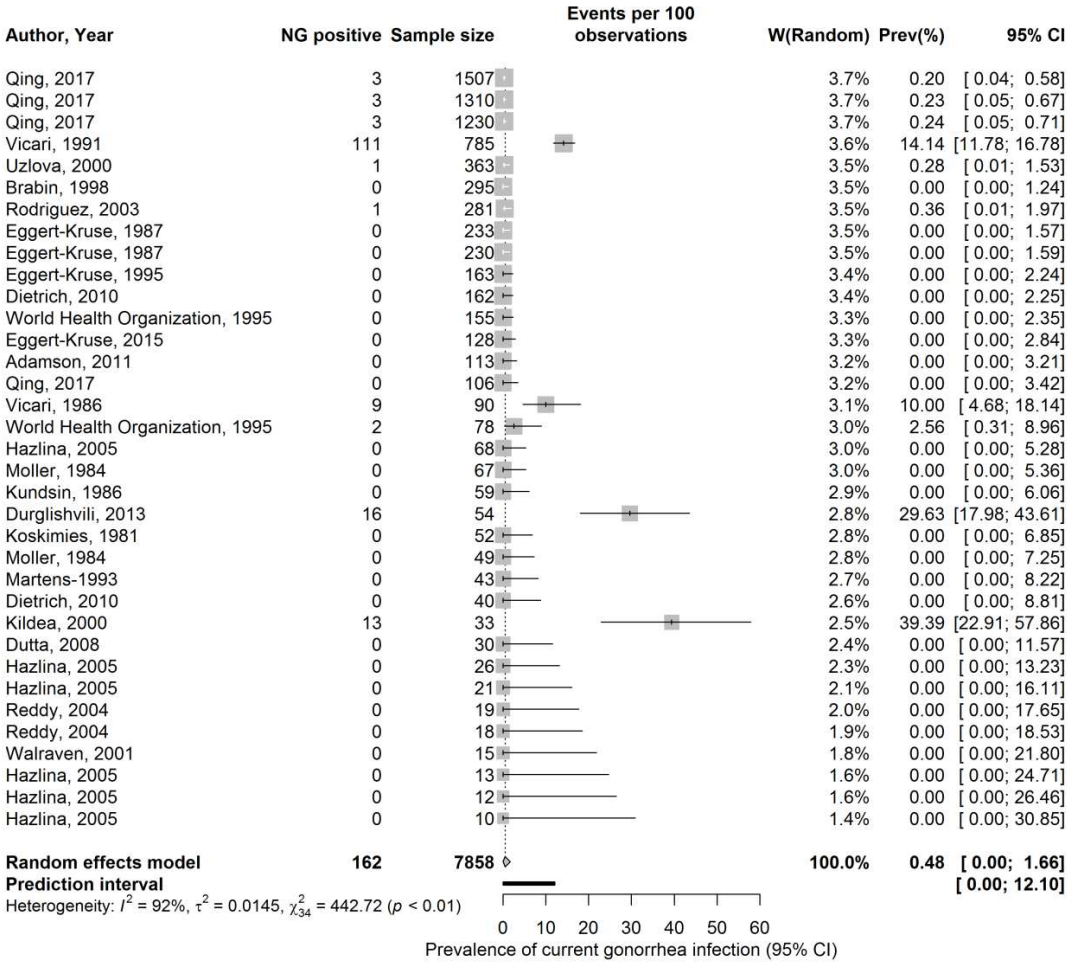
Supplementary Figure 4 Forest plots showing the results of meta-analysis on studies reporting the prevalence of current *Neisseria gonorrhoeae* (NG) infection in infertile populations as assessed using A) nucleic acid amplification assays, B) gram stain, and C) other assays assessing current infection



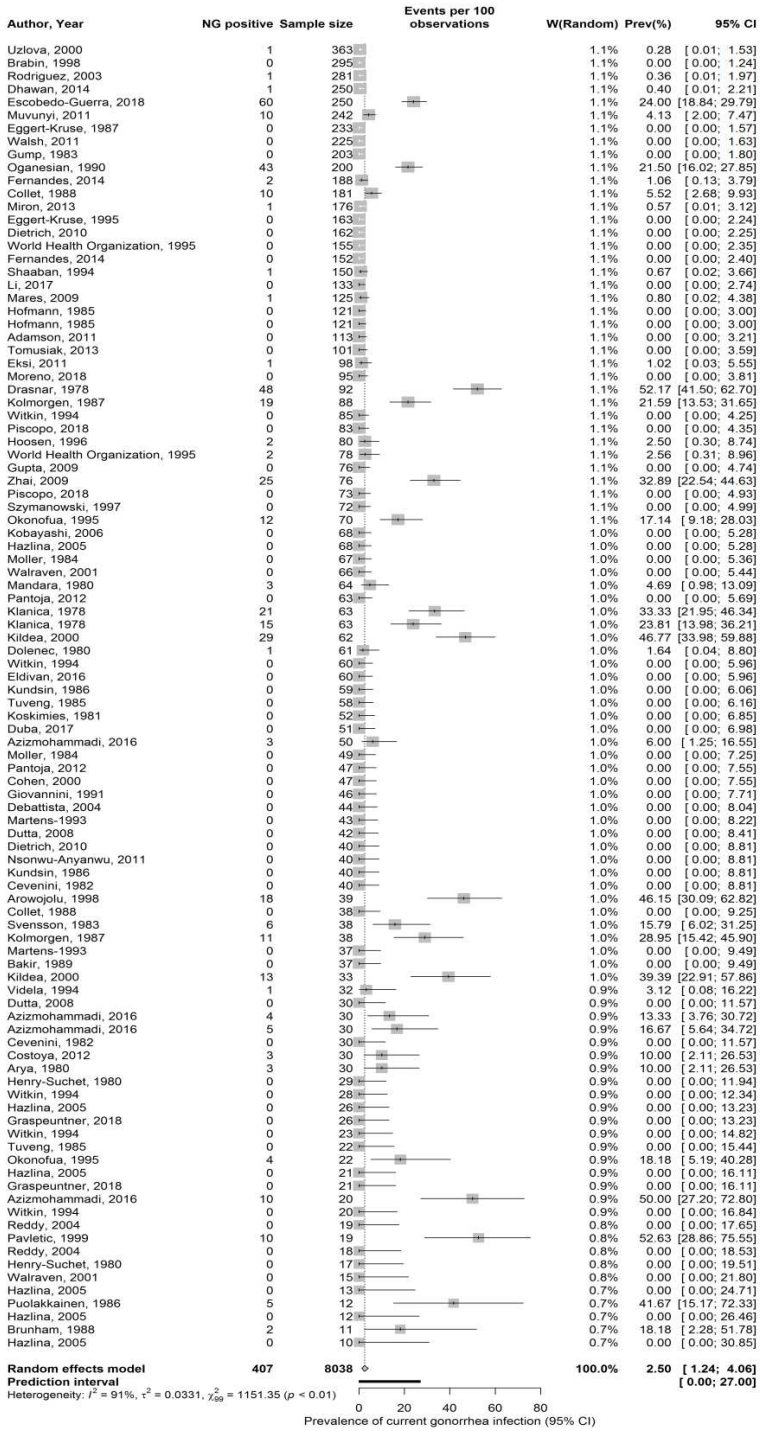
Supplementary Figure 5 Forest plots showing the results of meta-analysis on studies reporting the prevalence of current *Neisseria gonorrhoeae* (NG) infection in infertile populations in individuals with A) mixed and unexplained infertility, B) male factor infertility, and C) ovarian and non-tubal infertility



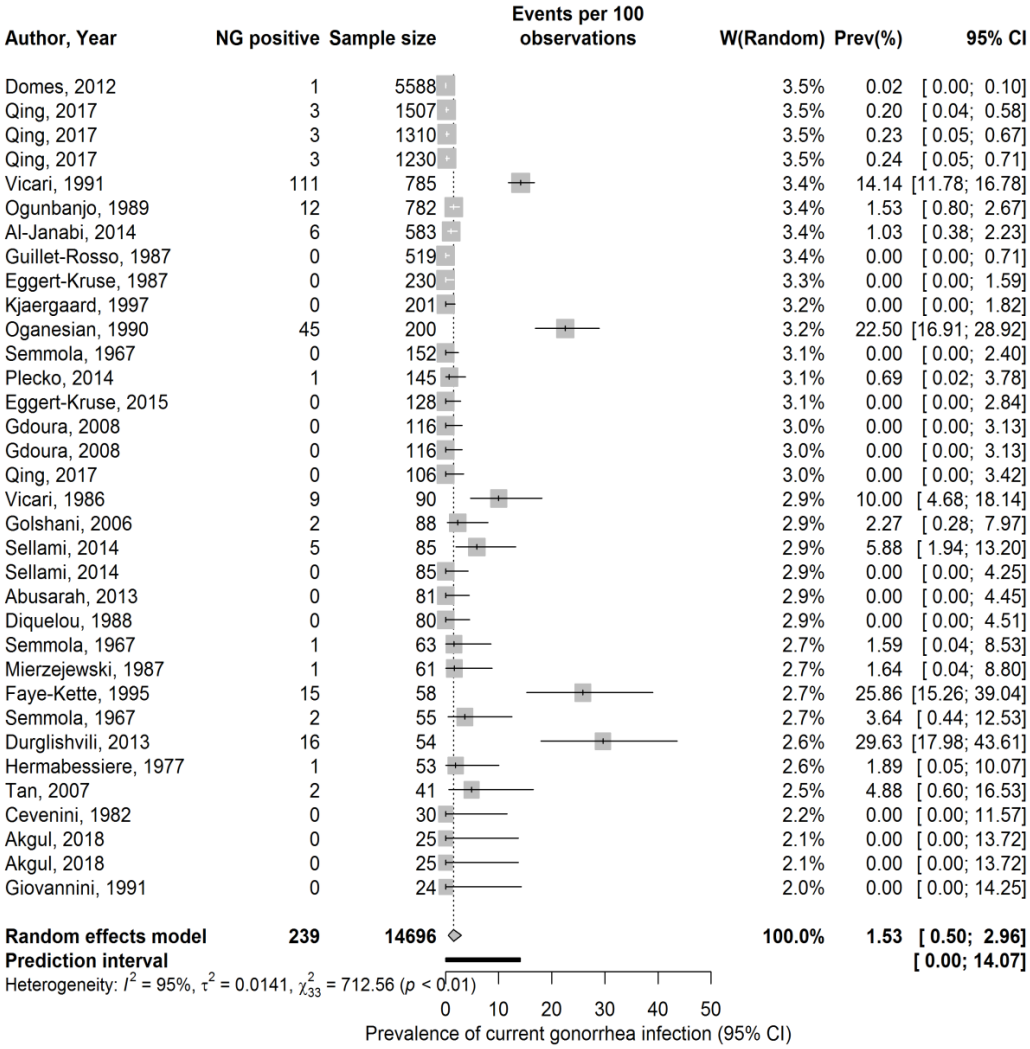
Supplementary Figure 6 Forest plot showing the results of meta-analysis on studies reporting the prevalence of current *Neisseria gonorrhoeae* (NG) infection in individuals with primary infertility



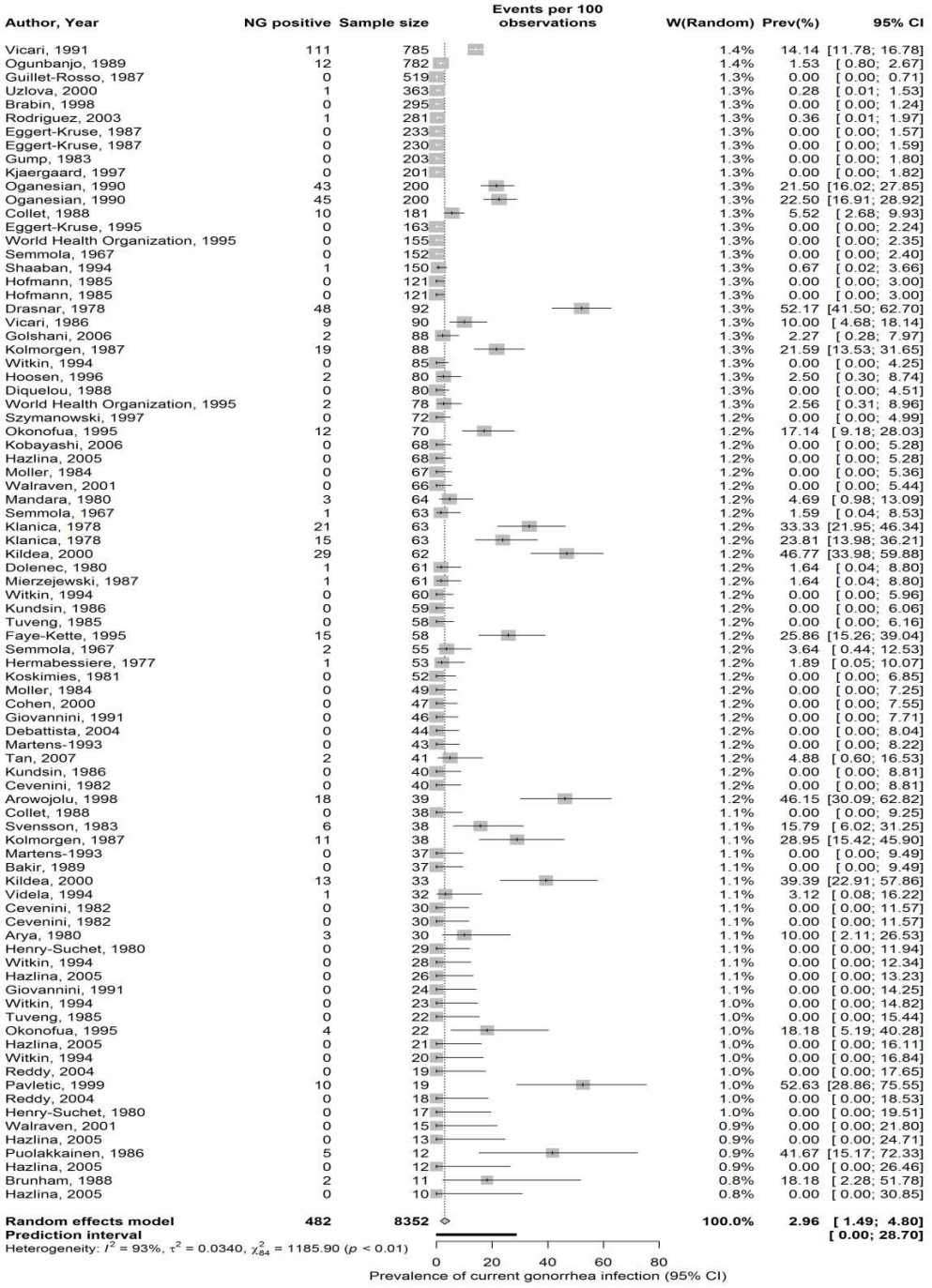
Supplementary Figure 7 Forest plot showing the results of meta-analysis on studies reporting the prevalence of current *Neisseria gonorrhoeae* (NG) infection in infertile women



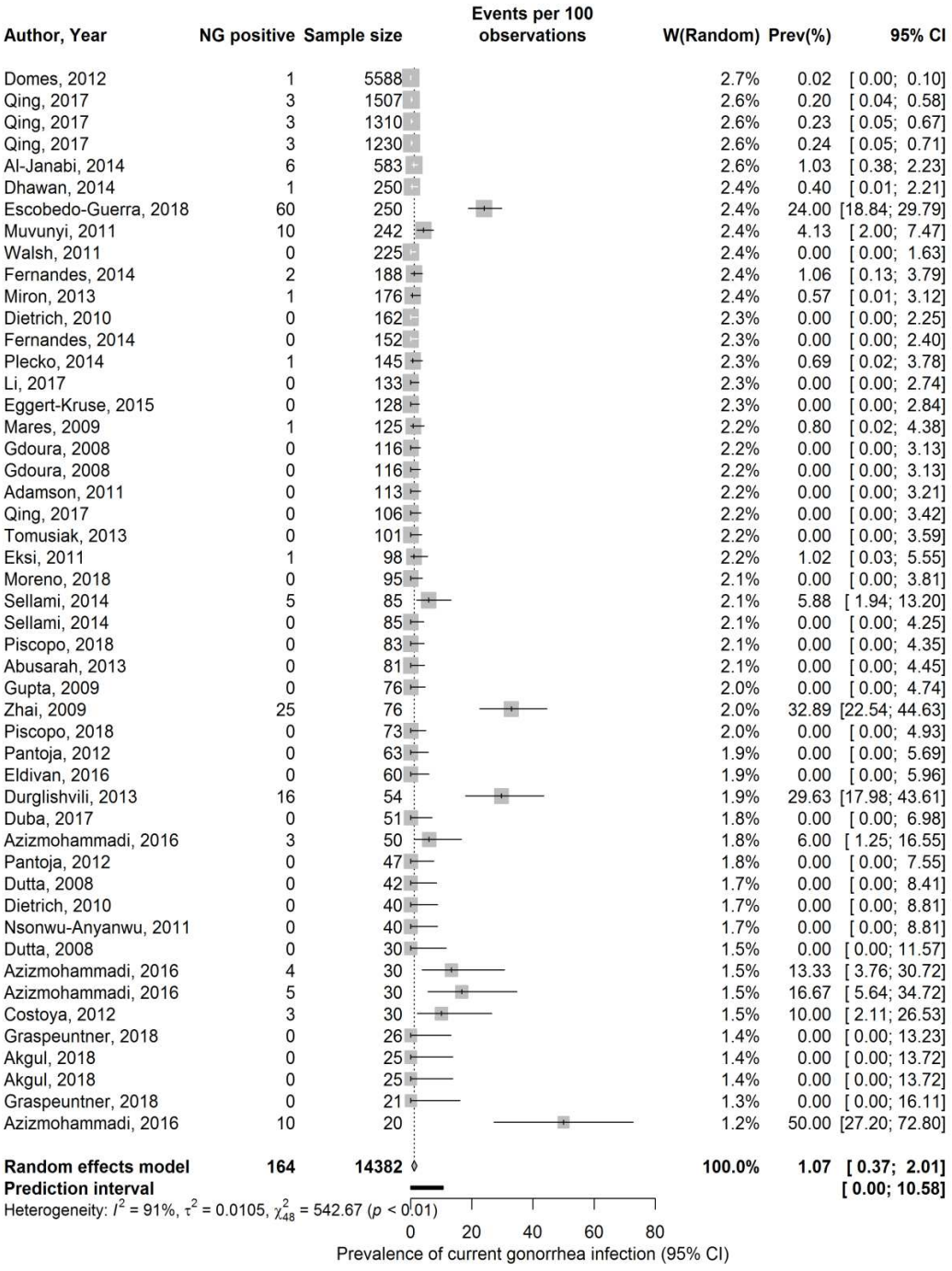
Supplementary Figure 8 Forest plot showing the results of meta-analysis on studies reporting the prevalence of current *Neisseria gonorrhoeae* (NG) infection in infertile men



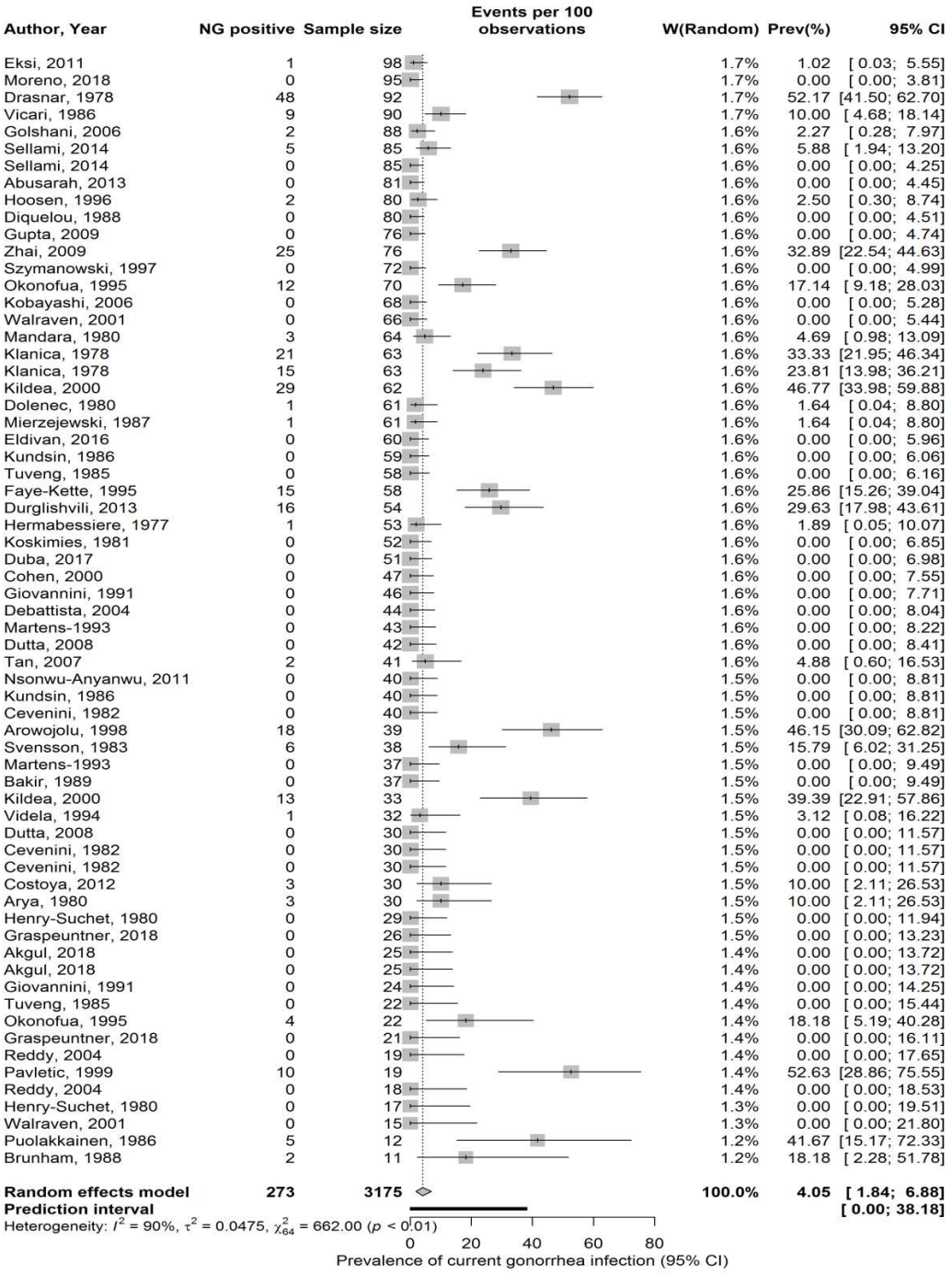
Supplementary Figure 9 Forest plot showing the results of meta-analysis on studies reporting the prevalence of current *Neisseria gonorrhoeae* (NG) infection in infertile populations before the year 2005



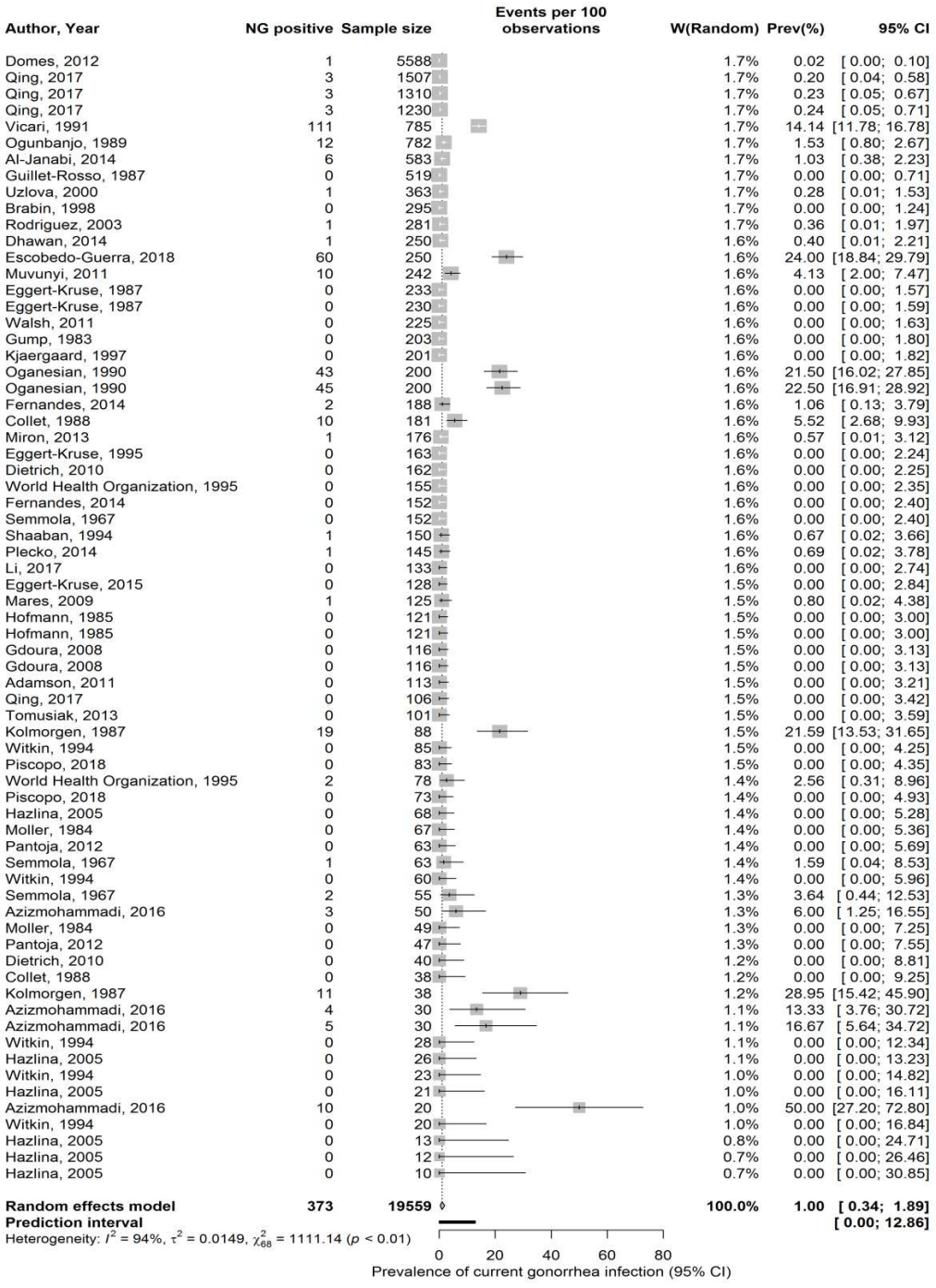
Supplementary Figure 10 Forest plot showing the results of meta-analysis on studies reporting the prevalence of current *Neisseria gonorrhoeae* (NG) infection in infertile populations from the year 2005 onwards



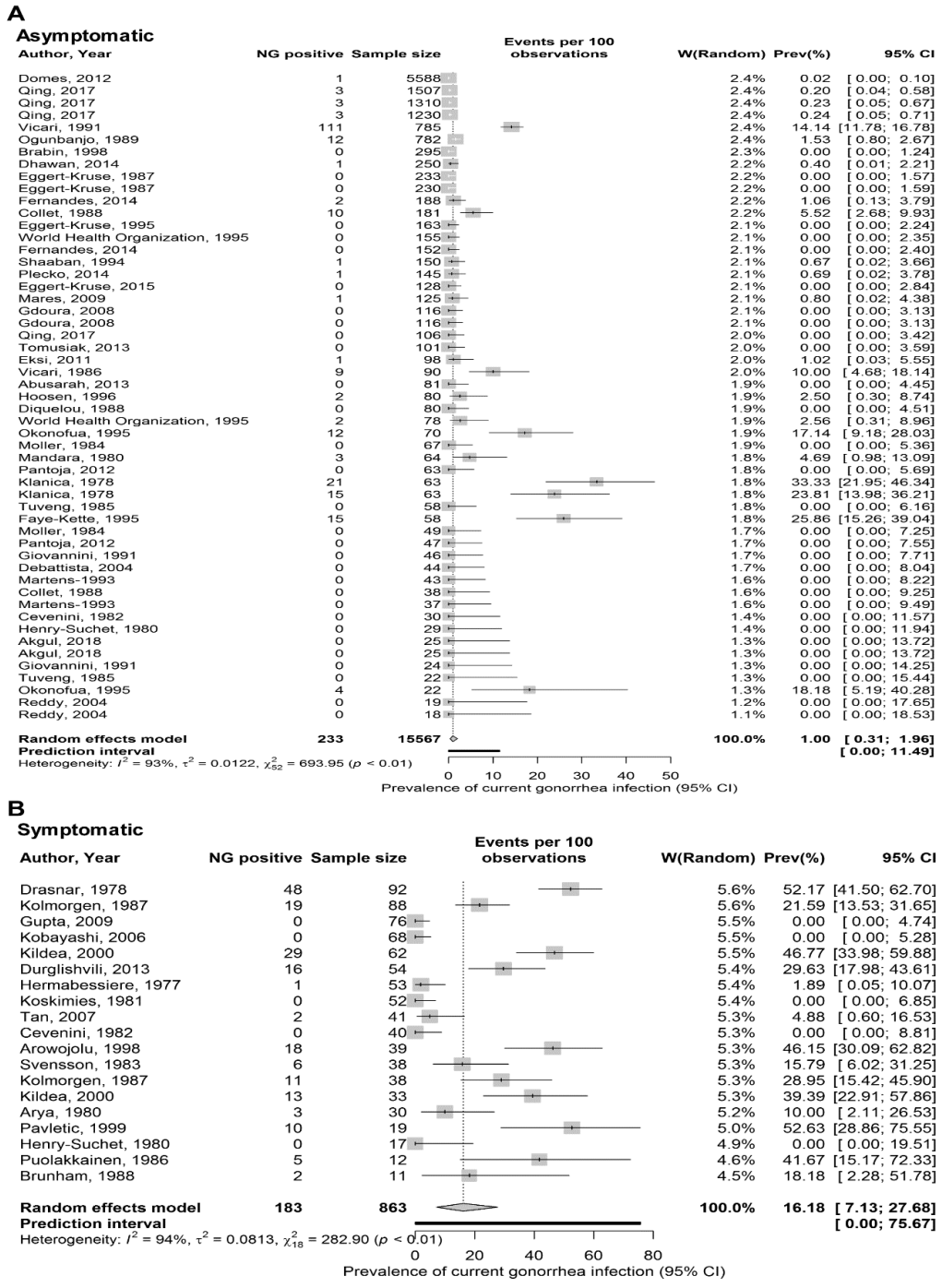
Supplementary Figure 11 Forest plot showing the results of meta-analysis on studies reporting the prevalence of current *Neisseria gonorrhoeae* (NG) infection in infertile populations using samples including fewer than 100 participants



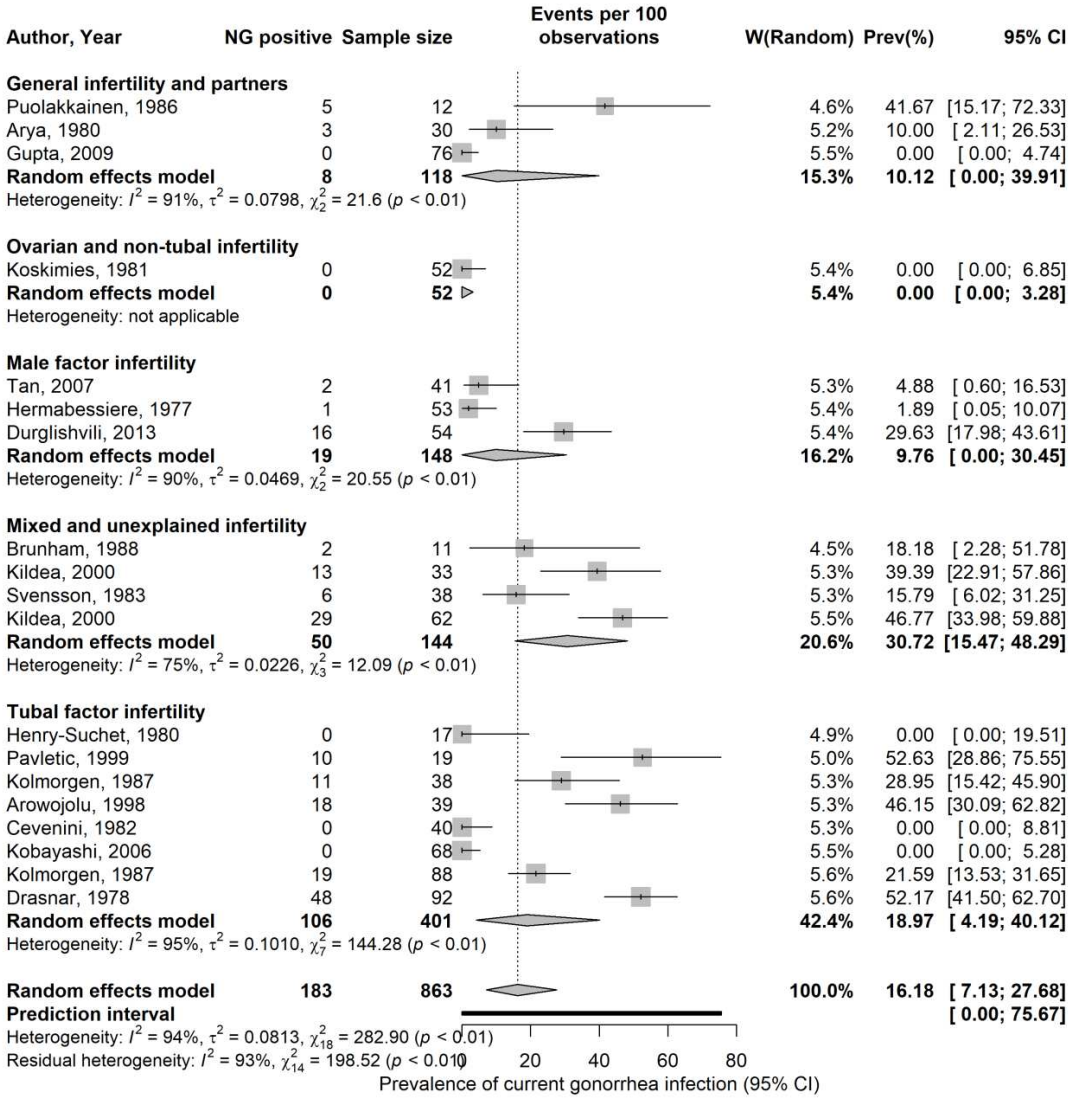
Supplementary Figure 12 Forest plot showing the results of meta-analysis on studies reporting the prevalence of current *Neisseria gonorrhoeae* (NG) infection in infertile populations using samples including 100 or more participants



Supplementary Figure 13 Forest plots showing the results of meta-analysis on studies reporting the prevalence of current *Neisseria gonorrhoeae* (NG) infection in infertile populations by absence (A) and presence (B) of urogenital signs and symptoms



Supplementary Figure 14 Forest plot showing the results of meta-analysis on studies reporting the prevalence of current *Neisseria gonorrhoeae* (NG) infection in symptomatic infertile populations by infertility diagnosis



Supplementary Table 11 Sensitivity analysis showing the results of univariable meta-regression analyses for the prevalence of current *Neisseria gonorrhoeae* (NG) infection in infertile populations

	Predictors		Studies/strata	Samples	Univariable analyses		Variance explained
			Total N	Total n	OR (95% CI)	p-value*	R^2 (%)
Population characteristics	**WHO region**	AMRO	24	7,204	1.00		0.9
		AFRO	16	1,825	2.96 (1.00-8.76)	0.050	
		EMRO	13	1,471	1.72 (0.54-5.47)	0.357	
		EURO	54	6,398	1.21 (0.53-2.75)	0.654	
		SEARO	8	843	0.55 (0.14-2.19)	0.397	
		WPRO	17	4,760	1.41 (0.48-4.09)	0.526	
		Multicenter	2	233	0.64 (0.05-7.60)	0.721	
	Sex	Men	34	14,696	1.00		1.7
		Women	100	8,038	1.84 (0.95-3.57)	0.073	
	Infertility type	Primary/Majority primary	35	7,858	1.00		2.1
		Secondary/Majority secondary	16	1,212	1.97 (0.72-5.40)	0.186	
		Not specified/Not applicable	83	13,664	2.09 (1.06-4.09)	0.033	
	Infertility diagnosis	General infertility/Not specified	55	11,136	1.00		2.4
		Tubal factor infertility	29	2,145	1.59 (0.74-3.42)	0.232	
		Male factor infertility	19	7,208	0.72 (0.32-1.86)	0.554	
		Ovarian & non-tubal infertility	17	1,709	0.50 (0.20-1.26)	0.140	
		Mixed & unexplained infertility	14	536	1.82 (0.67-4.94)	0.238	
	Presence of urogenital signs & symptoms	Asymptomatic	53	15,567	1.00		19.8
		Symptomatic	19	863	10.50 (4.67-23.58)	<0.001	
		Not specified	62	6,304	1.32 (0.75-2.32)	0.338	
Study methodology characteristics	**Assay type**	NAAT (current infection)	33	12,594	1.00		4.9
		Culture (current infection)	94	9,682	2.54 (1.31-4.96)	0.006	
		Other† (current infection)	7	458	4.08 (1.04-16.08)	0.044	
	Year of data collection‡		134	22,734	0.98 (0.96-1.00)	0.037	2.5
	Sample size	<100 participants	65	3,175	1.00		10.8
		≥100 participants	69	19,559	0.32 (0.18-0.55)	<0.001	
Risk of bias domains	**Infertility definition**	Follows WHO definition	50	9,397	1.00		0
		Otherwise/Unclear	84	13,337	1.30 (0.71-2.38)	0.389	
	Exposure to antimicrobials	Lack of exposure last week	23	1,954	1.00		0.4
		Exposure in last week	8	1,443	0.59 (0.15-2.35)	0.452	
		Unclear	103	19,337	1.42 (0.65-3.09)	0.372	
	Infection ascertainment	Consistency in assay used	132	22,644	1.00		0
		Otherwise/Unclear	2	90	1.57 (0.14-17.57)	0.712	
	Response rate	≥80%	2	70	1.00		0
		<80%/Unclear	132	22,664	0.52 (0.05-5.80)	0.592	

AFRO: African Region; AMRO: Region of the Americas; AOR: Adjusted odds ratio; CI: confidence interval; EMRO: Eastern Mediterranean Region; EURO: European Region; NAAT: nucleic acid amplification test; OR: Odds ratio; SEARO: South-East Asia Region; WHO: World Health Organization; WPRO: Western Pacific Region.

*Strength of evidence for an association with prevalence was deemed 'good' at 0.05< p-value ≤0.10, and 'strong' at p-value ≤0.05.
†Includes gram stain, gonozyme, and fluorescent antibody assays.
‡Missing values for year of data collection were imputed using data for year of publication adjusted by the median difference between year of publication and year of data collection (for studies with complete information).

Supplementary Table 12 Results of univariable and multivariable meta-regression analyses for the prevalence of *Neisseria gonorrhoeae* (NG) infection in infertile populations. Final multivariable meta-regression model reached through backward variable selection

	Predictors		Studies /strata	Samp-les	Univariable analyses			Variance explained	Multivariable analysis*		
			Total N	Total n	OR (95% CI)	p-value	LR test p-value†	R^2 (%)	AOR (95% CI)	p-value	LR test p-value‡
Population characteristics	**WHO region**	AMRO	33	7,960	1.00		0.018	5.1	1.00		0.071
		AFRO	24	2,281	3.95 (1.54-10.09)	0.004			2.05 (0.90-4.69)	0.088	
		EMRO	14	1,550	1.37 (0.45-4.19)	0.576			1.52 (0.58-3.94)	0.391	
		EURO	73	8,987	1.32 (0.64-2.76)	0.452			1.04 (0.56-1.95)	0.902	
		SEARO	15	1,207	1.24 (0.42-3.70)	0.692			0.55 (0.21-1.41)	0.211	
		WPRO	19	4,905	1.23 (0.45-3.37)	0.686			1.25 (0.53-2.95)	0.609	
		Multicenter	6	374	8.96 (1.90-42.29)	0.006			3.80 (0.90-15.95)	0.068	
	Sex	Men	43	16,560	1.00		0.031	2.0	--	--	--
		Women	141	10,704	1.98 (1.06-3.67)	0.031			--	--	--
	Infertility type	Primary/Majority primary	52	9,330	1.00		0.194	0.7	--	--	--
		Secondary/Majority secondary	24	1,756	2.14 (0.88-5.17)	0.091			--	--	--
		Not specified/Not applicable	108	16,178	1.06 (0.58-1.95)	0.840			--	--	--
	Infertility diagnosis	General infertility/Not specified	64	12,441	1.00		0.013	4.7	--	--	--
		Tubal factor infertility	50	3,238	2.39 (1.23-4.63)	0.010			--	--	--
		Male factor infertility	24	8,263	0.72 (0.31-1.67)	0.442			--	--	--
		Ovarian & non-tubal infertility	26	2,255	0.80 (0.36-1.81)	0.595			--	--	--
		Mixed & unexplained infertility	20	1,067	2.03 (0.83-4.98)	0.122			--	--	--
	Presence of urogenital signs & symptoms	Asymptomatic	64	16,484	1.00		<0.001	8.2	1.00		<0.001
		Symptomatic	24	1,408	5.94 (2.61-13.54)	0.000			4.98 (2.31-10.78)	<0.001	
		Not specified	96	9,372	1.67 (0.96-2.91)	0.070			1.48 (0.89-2.46)	0.133	
Study methodology characteristics	**Assay type**	NAAT (current infection)	33	12,594	1.00		<0.001	22.0	1.00		<0.001
		Culture (current infection)	94	9,682	2.54 (1.34-4.83)	0.005			1.76 (0.94-3.29)	0.076	
		Other§ (current infection)	7	458	4.08 (1.09-15.27)	0.037			1.97 (0.55-7.02)	0.294	
		IgG (Ever infection)	25	1,470	22.10 (9.53-51.20)	<0.001			17.40 (7.29-41.52)	<0.001	
		IgA/Unclear	25	3,060	3.74 (1.61-8.67)	0.002			2.94 (1.30-6.65)	0.010	
	Year of data collection£		184	27,264	0.97 (0.95-0.99)	0.001	0.001	5.5	--	--	--
	Sample size	<100 participants	81	3,803	1.00		0.011	3.0	1.00	--	0.031
		≥100 participants	103	23,461	0.51 (0.30-0.86)	0.011			0.58 (0.36-0.95)	0.031	--
Risk of bias domains	**Infertility definition**	Follows WHO definition	70	11,293	1.00		0.549	0	--	--	--
		Otherwise/Unclear	114	15,971	0.85 (0.49-1.46)	0.549			--	--	--
	Exposure to antimicrobials	Lack of exposure last week	25	2,037	1.00		0.048	2.2	--	--	--
		Exposure in last week	9	1,503	0.91 (0.23-3.61)	0.890			--	--	--
		Unclear	150	23,724	2.27 (1.06-4.90)	0.036			--	--	--
	Infection ascertainment	Consistency in assay used	179	26,612	1.00		0.750	0	--	--	--
		Otherwise/Unclear	5	652	1.30 (0.25-6.66)	0.750			--	--	--
	Response rate	≥80%	2	70	1.00		0.849	0	--	--	--

	<80%/Unclear	182	27,194	0.78 (0.06-10.10)	0.849	--	--	--

AFRO: African Region; AMRO: Region of the Americas; AOR: Adjusted odds ratio; CI: confidence interval; EMRO: Eastern Mediterranean Region; EURO: European Region; Ig: immunoglobulin; LR: likelihood ratio; NAAT: nucleic acid amplification test; OR: Odds ratio; SEARO: South-East Asia Region; WHO: World Health Organization; WPRO: Western Pacific Region.

*Adjusted R^2 in the final multivariable model=33.8%.

†Predictors with p-value ≤0.2 were eligible for inclusion in the multivariable analysis.

‡Predictors were retained in the model if the p-value for the association with prevalence was ≤0.1.

§Includes gram stain, gonozyme, and fluorescent antibody assays.

£Missing values for year of data collection were imputed using data for year of publication adjusted by the median difference between year of publication and year of data collection (for studies with complete information).

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